On the Decidability of $\mathcal{ALCI}_{RCC5}$: A Survey of the Problem, Its History, and a Proposed Solution

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Abstract

The description logic $\mathcal{ALCI}_{RCC5}$ extends $\mathcal{ALCI}$ with the five RCC5 base relations as roles, under composition-table semantics. Its decidability has been open since Wessel (2002/2003). We survey the history of the problem—from Cohn's original proposal (1993) through Wessel's systematic investigation and Lutz & Wolter's related undecidability results (2006)—and explain why all known undecidability reductions fail for $\mathcal{ALCI}_{RCC5}$. We then present a proposed decision procedure: the *cover-tree tableau*, which decomposes models into PPI-oriented cover trees where only $\{\mathsf{DR}, \mathsf{PO}\}$ remain as open cross-edges. This decomposition, combined with the patchwork property of RCC5, yields finite blocking via rank-d signatures. The approach is supported by extensive computational evidence (911 test concepts, zero mismatches; 775/775 SAT concepts with cover-tree models) but has not been formally verified by human experts. We discuss why naïve tableau approaches fail for $\mathcal{ALCI}_{RCC5}$ and how the cover-tree tableau overcomes these obstacles.

Keywords

description logic, spatial reasoning, RCC5, decidability, tableau calculus, patchwork property

1. Introduction

The decidability of concept satisfiability in the spatially extended description logic $\mathcal{ALCI}_{RCC5}$ has been an open problem for over twenty years. This paper provides a self-contained overview of the problem, its intellectual history, the obstacles that have resisted solution, and a proposed decision procedure that—if correct—would settle the question positively.

The paper is organized as follows. Section 2 traces the history of $\mathcal{ALCI}_{RCC5}$ from Cohn's original 1993 proposal through Wessel's systematic investigation and Lutz & Wolter's topological undecidability results. Section 3 explains why standard undecidability reductions fail, including a concrete domino-encoding attempt that we verify is blocked. Section 4 discusses why naïve tableau approaches fail for complete-graph logics. Section 5 presents the split-forest model decomposition. Section 6 describes the cover-tree tableau and explains how it achieves blocking. Section 8 concludes with an assessment of what has been achieved and what remains to be done.

Disclaimer and AI-tool disclosure. This paper was prepared by Michael Wessel with significant assistance from generative-AI language tools: Claude (Anthropic, Opus 4.7 and Opus 4.8) [48], GPT-5.4 Pro (OpenAI) [49], and GPT-5.5 (OpenAI) [16]. The AI tools were used to formalise the mathematical proofs, implement the cover-tree tableau decision procedure, conduct the computational verification campaigns, perform the cold reviews that found successive defects, and draft the manuscript text including this overview. Wessel reviewed and edited the content as needed, directed the research, identified and corrected errors, and takes full responsibility for the publication's content—including any errors, mistakes, or misleading statements the AI tools may have introduced. See the Declaration on Generative AI at the end of the paper for further details. The results and proofs have **not been peer-reviewed or verified by human domain experts**; they are published as a discussion piece, and the one completeness gap a cold review found (D-1) is repaired at the manuscript level in round-9 (§7.9) but not independently certified. The claims should not be taken as established results unless independently verified. Scrutiny, corrections, and feedback are invited.

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2. History of $\mathcal{ALCI}_{RCC5}$

2.1. Cohn's 1993 proposal

The idea of combining description logics with qualitative spatial reasoning via modal accessibility relations was first proposed by Cohn [1], who considered treating the RCC8 relations (or subsets thereof) as modalities in a multi-modal logic. The framework allowed spatial constraints to be expressed as modal formulas—e.g., $\langle \mathsf{PP} \rangle C$ for "some proper part satisfies C"—but Cohn left all decidability and complexity questions open.

2.2. Wessel's $\mathcal{ALCI}_{RCC}$ family (2002/2003)

Wessel [7, 8, 5] gave the first rigorous treatment of the combination, introducing the $\mathcal{ALCI}_{RCC}$ *family*: description logics $\mathcal{ALCI}_{RCCk}$ ($k = 1, 2, 3, 5, 8$) extending $\mathcal{ALCI}$ with role boxes derived from the RCC composition tables. In an $\mathcal{ALCI}_{RCCk}$ interpretation, every pair of domain elements is assigned exactly one RCCk base relation, and the composition table is enforced globally (*strong* EQ *semantics*: EQ is identity). Models are complete graphs with RCCk edge-colorings.

Wessel established the following:

- **Decidable cases:** $\mathcal{ALCI}_{RCC1}$, $\mathcal{ALCI}_{RCC2}$, and $\mathcal{ALCI}_{RCC3}$ are decidable, because their composition tables are deterministic or near-deterministic, admitting reduction to standard DL techniques.
- **Lower bounds:** $\mathcal{ALCI}_{RCC5}$ is PSPACE-hard; $\mathcal{ALCI}_{RCC8}$ is EXPTIME-hard.
- **Open:** The decidability of $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$ was left as the central open problem.

In a series of companion reports, Wessel also mapped out the decidability landscape for description logics with general composition-based role inclusion axioms:

- $\mathcal{ALC}_{\mathcal{RA}}$ (arbitrary role axioms) is **undecidable**, via reduction from the Post Correspondence Problem [6].
- $\mathcal{ALC}_{\mathcal{RA}}^{\ominus}$ (non-disjoint-roles variant) is also **undecidable**, via reduction from non-empty CFG intersection [3].
- $\mathcal{ALC}_{\mathcal{RA}_{SG}}$ (admissible role boxes: deterministic, functional, complete, and associative) is **decidable** (EXPTIME-complete, via reduction to $\mathcal{ALC}$ with general TBoxes) [4]; adding unqualified number restrictions ($\mathcal{ALCN}_{\mathcal{RA}_{SG}}$) makes it undecidable again.

$\mathcal{ALCI}_{RCC5}$ sits precisely in the gap: its role box is non-deterministic (unlike $\mathcal{ALC}_{\mathcal{RA}_{SG}}$) but has the patchwork property (unlike $\mathcal{ALC}_{\mathcal{RA}}/\mathcal{ALC}_{\mathcal{RA}}^{\ominus}$).

Remark 2.1 (Key difficulties)**.** $\mathcal{ALCI}_{RCC5}$ has none of the properties that standard DL decidability proofs rely on: (i) no tree model property (models are complete graphs), (ii) no finite model property (some satisfiable concepts require infinite models), (iii) a non-deterministic role box (multi-valued composition entries).

For reference, the full RCC5 composition table is given in Table 1. Reading: row = $S(b, c)$, column = $R(a, b)$, entry = possible relations for (a, c).
Key composition facts exploited in the cover-tree theory (Sections 5–6): $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ and $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$ (DR rigidity); $\mathsf{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$ and $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$ (transitivity); and $\mathsf{comp}(\mathsf{DR}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$ (PP can be *forced* by composition and universal filtering; see Example 6.2).

Table 1
The RCC5 composition table ($*$ = all five relations).

comp	DR(a,b)	PO(a,b)	EQ(a,b)	PPI(a,b)	PP(a,b)
DR(b,c)	$*$	DR, PO, PPI	DR	DR, PO, PPI	DR
PO(b,c)	DR, PO, PP	$*$	PO	PO, PPI	DR, PO, PP
EQ(b,c)	DR	PO	EQ	PPI	PP
PP(b,c)	DR, PO, PP	PO, PP	PP	PO, EQ, PP, PPI	PP
PPI(b,c)	DR	DR, PO, PPI	PPI	PPI	$*$

2.3. Wessel's grid constructions and the coincidence obstruction

In his investigation of $\mathcal{ALCI}_{RCC8}$ [8], Wessel made two important discoveries:

1. Even-odd chains (Section 4.4.2). The TPP/NTPP distinction in RCC8 allows the construction of *functional successor chains*: each node alternates between $even$ and odd, and a node can distinguish its direct TPPI-successor from all indirect ones by parity. Wessel used this to encode a binary n-bit counter, proving $\mathcal{ALCI}_{RCC8}$ is ExpTime-hard.

2. The grid coincidence obstruction (Section 4.4.3, Figure 11). Wessel constructed explicit RCC8 networks isomorphic to $n \times n$ grids, using TPP for horizontal and vertical successors. However, he identified a fundamental obstacle to encoding domino tilings: the *coincidence condition* $R_X \circ R_Y = R_Y \circ R_X$ (east-of-north equals north-of-east) cannot be enforced by concept terms. The composition table allows $\{PO, EQ, TPP, NTPP, TPPI, NTPPI\}$ at the coincidence point—too many options. Wessel concluded:

> "Even though these infinite models do exist, we have found no way to *enforce* the depicted model(s) by means of a concept term."

Wessel further observed that adding a hybrid logic binding operator $\downarrow$ (nominals) would immediately yield undecidability, confirming that the coincidence condition is the *precise boundary*.

Wessel thus independently developed, three to four years before Lutz–Wolter, the RCC8 description-logic analogues of these ingredients—the even-odd chain, an explicit two-dimensional TPP/NTPP grid, and the first identification of the coincidence gap as the decisive obstacle. Lutz–Wolter later obtained undecidability by a *different* route—a linearized tiling enumeration (their §5, following Marx–Reynolds) that sidesteps the two-dimensional coincidence rather than enforcing it—so this is shared lineage and prior identification, not a dependence of their proof on Wessel's constructions.

2.4. Lutz & Wolter (2006)

Lutz and Wolter [9] studied *modal logics of topological relations*, denoted L_{RCC8}, L_{RCC5}, etc. These have the same *syntax* as $\mathcal{ALCI}_{RCC}$ but are interpreted over *topological models*: the domain elements are regular closed subsets of a topological space (typically $\mathbb{R}^n$). They proved:

- L_{RCC8} is **undecidable** (their §5: a linearized NTPP-chain enumeration of tile positions, following Marx–Reynolds, whose faithfulness relies on the geometry of $\mathbb{R}^n$; no two-dimensional grid coincidence is enforced).
- $L_{RCC5}(RS^{\exists})$ is **undecidable** (reduction from $S5^3$ via supremum regions).
- $L_{RCC5}(RS)$ is **open**. Lutz and Wolter explicitly wrote (p. 31): "Perhaps the most interesting candidate is $L_{RCC5}(RS)$ [...] to which the reduction exhibited in Section 8 does not apply."

The logic $\mathcal{ALCI}_{RCC5}$ is exactly $L_{RCC5}(RS)$: arbitrary complete-graph models with composition-table semantics. The problem has been recognized as open by leading researchers since 2006.

Table 2
Why standard undecidability reductions fail for $\mathcal{ALCI}_{RCC5}$.

Source Problem	**Technique**	**Missing in $\mathcal{ALCI}_{RCC5}$**
$\mathbb{N} \times \mathbb{N}$ domino	Graded modalities + transitivity	Number restrictions
$\mathcal{ALC}_{\mathcal{RA}}$ [6]	Post Correspondence Problem	Arbitrary role axioms
$\mathcal{ALC}^{\ominus}_{\mathcal{RA}}$ [3]	Non-empty CFG intersection	Non-disjoint roles
$\mathcal{ALCI}_{RCC8}$ (Wessel)	Grid via TPP/NTPP	TPP/NTPP distinction
L_{RCC8} (Lutz–Wolter)	Linearized topological enumeration	Topological rigidity
$L_{RCC5}(RS^{\exists})$	$S5^3$ via supremum regions	Supremum closure

3. Why Undecidability Reductions Fail

Undecidability proofs for description logics use a variety of techniques: grid encoding ($\mathbb{N} \times \mathbb{N}$ domino tiling, requiring functional roles or number restrictions), reduction from the Post Correspondence Problem (PCP), reduction from context-free grammar (CFG) intersection, or simulation of $S5^3$ via geometric constructions. $\mathcal{ALCI}_{RCC5}$ lacks the features that each of these techniques exploits: it has no functional roles, no number restrictions, no arbitrary role box, and no geometric coincidence conditions.

3.1. Blocked reduction routes

Table 2 summarizes the blocked routes. The **patchwork property** of RCC5 (Renz & Nebel [2])—local (triple-wise) consistency implies global consistency for atomic networks—actively resists grid encoding, since tiling reductions require rigid global constraints that go beyond local consistency.

3.2. A concrete domino attempt in $\mathcal{ALCI}_{RCC5}$

Can we encode a domino-like grid using PP/PPI chains in RCC5? Consider the concept:

$$\begin{aligned} X \;=\; & \exists \mathsf{PPI}.\exists \mathsf{PPI}.C \;\sqcap\; \exists \mathsf{PPI}.\exists \mathsf{PPI}.D \\ & \sqcap\; \forall \mathsf{PPI}.\forall \mathsf{PPI}.(\forall \mathsf{PO}.(\neg C \sqcap \neg D) \sqcap \forall \mathsf{DR}.(\neg C \sqcap \neg D) \\ & \qquad\qquad \sqcap\; \forall \mathsf{PPI}.(\neg C \sqcap \neg D) \sqcap \forall \mathsf{PP}.(\neg C \sqcap \neg D)) \\ & \sqcap\; \forall \mathsf{PPI}.\exists \mathsf{PPI}.X \end{aligned}$$

The intent: each X-node has two PPI-grandchildren colored C and D (the "square"), and the universals isolate the colored nodes from all non-EQ neighbors. The recursive $\forall\mathsf{PPI}.\exists\mathsf{PPI}.X$ forces infinite repetition.

One level is satisfiable. Without the recursive conjunct, X has a 3-element model: z PP y PP x, with $z \in C \sqcap D$. The two grandchildren are forced to be *the same element* (EQ collapse): if $z_1 \neq z_2$, they would be related by some $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$, and the universals would require $z_2 \notin D$ (firing z_1's $\forall R.\neg C \sqcap \neg D$). So $z_1 = z_2 \in C \sqcap D$, and the universals are vacuously satisfied (the only neighbors of z are PP-predecessors, not PO/DR/PPI-neighbors).

Two levels is unsatisfiable. Adding $\forall\mathsf{PPI}.X$: the PPI-child y must itself satisfy X (without recursion), producing a new $C \sqcap D$-labeled element w as a PPI-child of z. But z is a PPI-grandchild of x, and x's universals fire: z's PPI-neighbor w must satisfy $\neg C \sqcap \neg D$. Yet $w \in C \sqcap D$. **Contradiction.**

This was verified computationally: a simplified version $X_1 = \exists\mathsf{PPI}.\exists\mathsf{PPI}.C \sqcap \forall\mathsf{PPI}^3.\neg C$ is SAT; $X_2 = X_1 \sqcap \forall\mathsf{PPI}.X_1$ is UNSAT. The domino encoding collapses at depth 2 because the isolation constraints from the grandparent level kill the colored elements created by the recursive unfolding.

Remark 3.1. The failure is not accidental. In $\mathcal{ALCI}_{RCC5}$, the only way to create "functional" successors is via PP/PPI chains, but $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$ makes these chains *transitive*—grandparent universals automatically propagate to grandchildren, preventing the positional diversity needed for grid encoding. In $\mathcal{ALCI}_{RCC8}$, the $\mathsf{TPP}/\mathsf{NTPP}$ distinction breaks this transitivity (a direct TPPI-child is distinguishable from indirect descendants), which is why Wessel's even-odd chain works there but has no RCC5 analogue.

3.3. A concrete domino attempt in $\mathcal{ALCI}_{RCC8}$

The remark above suggests the $\mathsf{TPPI}/\mathsf{NTPPI}$ distinction in $\mathcal{ALCI}_{RCC8}$ might rescue the construction: a direct successor (TPPI) and an indirect one (NTPPI) are genuinely distinguishable, so perhaps the level-2 collapse can be avoided. We attempt the analogue and show that while it escapes the RCC5 failure mode, it fails for a different reason—Wessel's *coincidence obstruction* [8].

The schema. Let Corner be the concept that isolates a node from all non-EQ neighbors:

$$\mathsf{Corner} \;=\; \bigwedge_{R\in\{\mathsf{DC},\mathsf{EC},\mathsf{PO},\mathsf{TPP},\mathsf{NTPP},\mathsf{TPPI},\mathsf{NTPPI}\}} \forall R.(\neg C \sqcap \neg D).$$

The isolation is placed *locally on the colored grandchildren* rather than as an ancestral $\forall\mathsf{TPPI}.\forall\mathsf{TPPI}$ universal, so it cannot propagate transitively through the recursion:

$$\begin{aligned} X \;&= \exists\mathsf{TPPI}.\exists\mathsf{TPPI}.(C \sqcap \mathsf{Corner}) \;\sqcap\; \exists\mathsf{TPPI}.\exists\mathsf{TPPI}.(D \sqcap \mathsf{Corner}) \\ &\quad \sqcap\; \forall\mathsf{TPPI}.(\neg C \sqcap \neg D) \;\sqcap\; \forall\mathsf{TPPI}.X. \end{aligned}$$

The third conjunct $\forall\mathsf{TPPI}.(\neg C \sqcap \neg D)$ forces the corner relation $x \to z$ to be NTPPI (not TPPI), using $\mathsf{comp}(\mathsf{TPPI}, \mathsf{TPPI}) = \{\mathsf{TPPI}, \mathsf{NTPPI}\}$.

Level 1 is satisfiable, with a real direct/indirect distinction. Without recursion, x has TPPI-children y_1, y_2 (the two "direct" positions) and a single corner z reached by TPPI-TPPI chains from both y_1 and y_2. The EQ-collapse at level 2 works exactly as in RCC5: $z_1 \in C \sqcap \mathsf{Corner}$ and $z_2 \in D \sqcap \mathsf{Corner}$ are mutual non-EQ neighbors only if Corner is violated, so under strong-EQ semantics $z_1 = z_2 = z$. Crucially, $z \neq y_1, y_2$ in general: z is NTPPI of x while y_1, y_2 are TPPI—the direct/indirect distinction survives, which is a genuine RCC8 refinement over the RCC5 attempt of §3.2.

Level 2 collapses via the coincidence obstruction. Adding $\forall\mathsf{TPPI}.X$: the TPPI-child y of x satisfies X, producing its own corner $w \in C \sqcap D \sqcap \mathsf{Corner}$ at NTPPI-distance from y, and hence at NTPPI-distance from x via $\mathsf{comp}(\mathsf{TPPI}, \mathsf{NTPPI}) = \{\mathsf{NTPPI}\}$. Now x's own corner z and y's corner w are distinct grid positions in the intended reading—but as elements of the model they are related by some RCC8 base relation. Since z carries Corner, every non-EQ neighbor of z lies in $\neg C \sqcap \neg D$. But $w \in C \sqcap D$. Therefore $z = w$ under strong-EQ. **All corners across the grid coincide at a single point.**

Why the $\mathsf{TPPI}/\mathsf{NTPPI}$ trick fails. In the RCC5 attempt the collapse is a *transitive-propagation* failure: the grandparent universal fires on great-grandchildren because $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$. Moving the isolation to a local Corner marker and using the non-transitive piece $\mathsf{comp}(\mathsf{TPPI}, \mathsf{TPPI}) = \{\mathsf{TPPI}, \mathsf{NTPPI}\}$ eliminates that specific pathway—the universal at the grandparent no longer reaches deeper nodes. But a *different* obstruction takes over: the composition table permits any of $\{\mathsf{PO}, \mathsf{EQ}, \mathsf{TPP}, \mathsf{NTPP}, \mathsf{TPPI}, \mathsf{NTPPI}\}$ between two corners at different intended grid coordinates, and nothing in the concept language rules out EQ. The Corner markers themselves then force the EQ-merge. This is exactly the obstruction Wessel identified in report7.pdf, Figure 11 (cf. §2.3): the composition table does not carry the rigidity needed to keep distinct grid cells distinct.

Remark 3.2. The failure mode is structurally different from the RCC5 one but traces back to the same root cause. Lutz–Wolter's L_{RCC8} reduction *circumvents* the coincidence obstruction rather than enforcing it: they linearize the plane into a single NTPP-chain (the Marx–Reynolds enumeration) and recover the vertical direction from the enumeration, so no "north-of-east = east-of-north" coincidence has to be forced; the geometry of $\mathbb{R}^n$ only has to keep that linear encoding faithful [9]. No such rigidity is available in the abstract RCC8 semantics of $\mathcal{ALCI}_{RCC8}$: a pure composition table allows corner points to merge freely, and the $\mathsf{TPPI}/\mathsf{NTPPI}$ distinction—while useful locally for a direct/indirect separation at level 2—cannot be chained to enforce an unbounded $\mathbb{N} \times \mathbb{N}$ grid. This matches the open-problem status of $\mathcal{ALCI}_{RCC8}$ decidability noted in [9] and further discussed in [8].

4. Why Naïve Tableau Blocking Fails

Standard DL tableau calculi achieve termination through *blocking*: if a node x has the same label as an ancestor y, we can reuse y's subtree for x, cutting off infinite branches. The key requirement is that the "reuse" preserves all constraints—in tree-shaped models, a blocked node's interactions are limited to its parent edge, so label equality suffices.

In $\mathcal{ALCI}_{RCC5}$, models are *complete graphs*: every pair of domain elements is related by an RCC5 base relation. A node interacts not just with its parent and children, but with *every other node in the model*. This creates a fundamental obstacle for blocking.

Example 4.1 (Blocking breaks with cross-edges)**.** Consider $C_0 = \exists\mathsf{PPI}.A \sqcap \exists\mathsf{DR}.\neg A \sqcap \exists\mathsf{PO}.\top$. A tableau might generate nodes:

- x_0: root, satisfies C_0
- x_1: PPI-child of x_0, type contains A
- x_2: DR-witness for x_0, type contains $\neg A$

Now x_1 and x_2 must be related by some RCC5 relation R, and R must be safe for their types. If x_1 generates a descendant x_3 with the same label as x_1, naïve blocking would declare x_3 blocked. But x_3's relation to x_2 may differ from x_1's relation to x_2: via composition, $\mathsf{comp}(\mathsf{inv}(\mathsf{PPI}), \mathsf{DR}) = \mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ forces x_1 to be DR-related to x_2, while x_3 (deeper in the tree) may face a different forced relation. The blocked node's cross-edge commitments differ from the blocking node's.

Multiple sophisticated blocking strategies have been tried and failed for $\mathcal{ALCI}_{RCC5}$ [47, 36, 35]:

1. **Profile-cached blocking** (GPT-5.4) [35]: Caches "coherent predecessor blocks" to compare global neighborhoods. Fails because the *color structure changes during unraveling*—when a blocked subtree is replayed, the edge-coloring to nodes outside the subtree is not preserved.

2. **Meet-semilattice replay** (GPT-5.4) [35]: Uses a meet-semilattice structure on labels to ensure robust normalization. Same unraveling gap.

3. **Tri-neighborhood blocking** (Claude) [32]: Matches not just node labels but entire *triangle neighborhoods* (the set of all types of 2-element subsets involving the node). **Termination disproved**: frontier nodes continuously acquire new triangle types as new neighbors are added, so $\mathrm{Tri}(x) \subsetneq \mathrm{Tri}(y)$ for frontier x and interior y—blocking never triggers.

4. **Finite-witness / patchwork-augmentation probes** (Claude) [31, 33, 34]: Probes of a finite-witness conjecture and typed-patchwork augmentations for the contextual tableau, both refuted by explicit counterexamples.

The common obstacle is the **blocking dilemma**: making the blocking condition strong enough for soundness (matching all cross-edge interactions) prevents termination (the cross-edge context keeps growing), and vice versa. A full catalogue of retracted approaches is maintained in [47]; the overall project is summarised at the repository [46].

5. Split-Forest Model Decomposition

The breakthrough insight, due to Wessel, is to decompose $\mathcal{ALCI}_{RCC5}$ models into a *tree layer* and a *cross-edge layer*, where the cross-edge layer is simple enough to handle without full global blocking.

Figure 1 reproduces the original whiteboard sketch from April 7, 2026: PPI chains form trees, DR/PO are sibling cross-edges, and DR propagates rigidly downward along PP-chains.

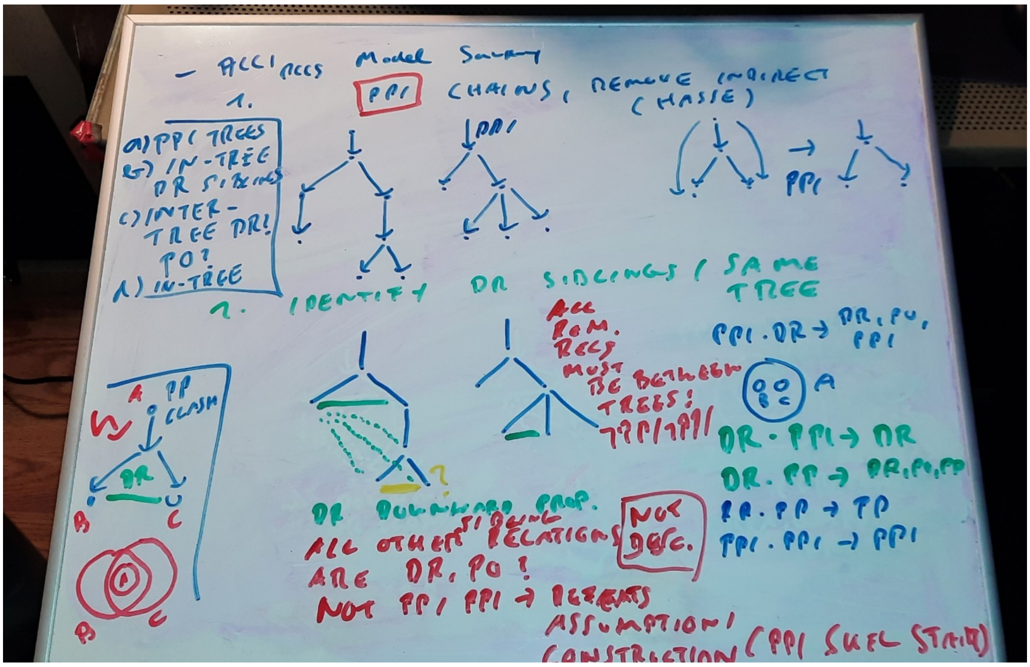

Figure 1: Wessel's whiteboard sketch (April 7, 2026) of the split-forest decomposition for $\mathcal{ALCI}_{RCC5}$. Vertical PPI-chains form the cover-tree backbone (with join nodes split into EQ-mates); DR/PO become sibling cross-edges between non-comparable subtrees; and DR propagates rigidly downward because $\mathrm{comp}(\mathsf{PP}, \mathsf{DR}) = \mathrm{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$, so the residual cross-edge alphabet is just $\{\mathsf{DR}, \mathsf{PO}\}$, which is trivially arc-consistent. Every subsequent formalisation in this paper—GPT-5.4's split-forest semantics, Claude's cover-tree tableau, and GPT-5.5's round-7-through-round-9 sketches—rests on this two-layer decomposition.

5.1. Cover trees

In any $\mathcal{ALCI}_{RCC5}$ model, the PP/PPI relation is a strict partial order (by composition: $\mathrm{comp}(\mathsf{PP}, \mathsf{PP}) = \{\mathsf{PP}\}$). Orient this order downward (PPI = "has proper part") to obtain a DAG. If the DAG has join nodes (elements with multiple PP-predecessors), split them into EQ-copies, one per parent path. The result is a rooted forest—the *cover tree*.

Definition 5.1 (Cover tree)**.** A *cover-tree presentation* of an $\mathcal{ALCI}_{RCC5}$ model $\mathcal{I}$ is a rooted tree T with a surjection $\mathrm{orig} : T \to \Delta^{\mathcal{I}}$ such that: (i) u is the parent of v in T iff $\mathrm{orig}(u)$ PPI $\mathrm{orig}(v)$ (immediate containment), and (ii) tree nodes mapping to the same element are EQ-mates (weak-EQ semantics, convertible to strong-EQ by typed congruence quotient).

5.2. The key observation: DR rigidity

In the cover tree, what relations can hold between nodes in different sibling subtrees? Consider sibling roots s and t (children of the same parent), and elements x below s, y below t in the tree. An element x below s is in the Core of the sibling pair (s, t) if $\mathrm{orig}(x)$ is a common descendant of both $\mathrm{orig}(s)$ and $\mathrm{orig}(t)$ in the original model's PP order—i.e., $\mathrm{orig}(x)$ is a proper part of both. Core elements are handled by EQ-splitting; the interesting question concerns non-Core elements.

Proposition 5.2 (Wessel)**.** *After* EQ*-splitting, only* DR *and* PO *are possible between non-*Core *elements of sibling subtrees.*

Proof.
- EQ(x, y): In the original DAG (before splitting), $x = y$ would mean a single element is a PP-descendant of both s and t—i.e., it is in the Core. After splitting, this element is duplicated

into distinct tree nodes x' below s and y' below t, with $\mathrm{orig}(x') = \mathrm{orig}(y')$ (EQ-mates in the weak-EQ presentation). Since $x' \neq y'$ as tree nodes, $\mathsf{EQ}(x', y')$ is impossible under strong EQ semantics.

- $\mathsf{PP}(x, y)$: would mean x lies below y in the part-of order, hence below t, making x a common descendant (Core).
- $\mathsf{PPI}(x, y)$: would place y below x below s, meaning t lies below s—contradicting sibling incomparability.

Only DR and PO remain. Furthermore, DR propagates rigidly: $\mathsf{comp}(\mathsf{PP}, \mathsf{DR}) = \{\mathsf{DR}\}$ and $\mathsf{comp}(\mathsf{DR}, \mathsf{PPI}) = \{\mathsf{DR}\}$, so if s DR t, then *every* descendant of s is DR to t and all its descendants. □

5.3. Trivial arc-consistency of $\{\mathsf{DR}, \mathsf{PO}\}$ networks

The consequence of Proposition 5.2 is that the cross-edge constraint network uses only $\{\mathsf{DR}, \mathsf{PO}\}$ as possible relations (plus $\{\mathsf{DR}\}$ for DR-rigid pairs).

Proposition 5.3. *For all $R, S \in \{\mathsf{DR}, \mathsf{PO}\}$: $\mathsf{comp}(R, S) \cap \{\mathsf{DR}, \mathsf{PO}\} \neq \varnothing$.*

This means a $\{\mathsf{DR}, \mathsf{PO}\}$ disjunctive network is *trivially arc-consistent*: every domain is either $\{\mathsf{DR}\}$, $\{\mathsf{PO}\}$, or $\{\mathsf{DR}, \mathsf{PO}\}$, and composing any two non-empty domains produces a non-empty result. By the patchwork property [2], the network therefore has a globally consistent atomic refinement whenever every domain is non-empty.

This is the central simplification: *checking that each type pair has a non-empty $\{\mathsf{DR}, \mathsf{PO}\}$-safe set suffices for cross-edge consistency.* No full path-consistency computation is needed.

6. The Cover-Tree Tableau

6.1. Algorithm overview

The cover-tree tableau searches for a *valid type set*: a finite set $T \subseteq \mathsf{Tp}(C_0)$ of Hintikka types that contains a root type realizing C_0 and satisfies four conditions:

(CT1) **Demand closure**: every existential demand $\exists R.D$ in every $\tau_i \in T$ has an R-safe witness $\tau_j \in T$ with $D \in \tau_j$.

(CT2) **Cross-edge consistency**: every pair $\tau_i, \tau_j \in T$ has either $\mathsf{Safe}(\tau_i, \tau_j) \cap \{\mathsf{PP}, \mathsf{PPI}\} \neq \varnothing$ (tree-relatable) or $\mathsf{Safe}(\tau_i, \tau_j) \cap \{\mathsf{DR}, \mathsf{PO}\} \neq \varnothing$ (cross-relatable).

(CT3) **Cover-tree sibling compatibility**: for each type τ_i with multiple DR/PO demands, the witnesses can be jointly assigned such that every witness pair satisfies $\mathsf{comp}(\mathsf{inv}(R_m), R_{m'}) \cap \{\mathsf{DR}, \mathsf{PO}\} \cap \mathsf{Safe}_{\mathsf{DR/PO}}(\tau_{j_m}, \tau_{j_{m'}}) \neq \varnothing$.

(CT4) **Composition propagation**: for each type with multiple demands of *any* role combination, witnesses τ_{j_1}, τ_{j_2} must satisfy $\mathsf{comp}(\mathsf{inv}(R_1), R_2) \cap \mathsf{Safe}(\tau_{j_1}, \tau_{j_2}) \neq \varnothing$.

The algorithm terminates because $|\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|}$ is finite, and the type set grows monotonically during the search.

6.2. How blocking works

The crucial difference from naïve tableau approaches is the *separation of layers*:

- The **tree layer** (PP/PPI) handles ancestor/descendant demands. Since PP/PPI form a tree, standard tree-model arguments apply: blocking via *rank-d signatures* (the type information visible within modal depth $d = \mathsf{md}(C_0)$) yields a finite representation.

- The **cross layer** (DR/PO) handles demands for disconnected or partially overlapping regions. Because the cross-edge network uses only $\{\mathsf{DR}, \mathsf{PO}\}$—which is trivially arc-consistent—the cross-layer check reduces to a simple pairwise non-emptiness condition (CT2), not a global consistency computation.

Naïve blocking fails because it tries to handle both layers simultaneously: the tree structure provides natural blocking, but the cross-edges to distant nodes keep changing, preventing a match. The cover-tree tableau sidesteps this by *not* blocking on cross-edge context—instead, it verifies cross-edge consistency as a separate, global check on the finite type set.

Example 6.1 (A concept that defeats naïve blocking but works with cover trees)**.** Consider $C_0 = \exists\mathsf{PO}.A \sqcap \exists\mathsf{PP}.\neg B \sqcap \forall\mathsf{DR}.A$. In a naïve tableau, the PP-child needs further expansion; blocking requires matching both the label *and* the triangle neighborhood (relations to the PO-witness and all other nodes). Since the PO-witness forces DR-relations to the PP-child's subtree (via $\mathsf{comp}(\mathsf{PO}, \mathsf{PP}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$), the cross-edge context keeps evolving.

In the cover-tree tableau: the PP-witness is an ancestor in the tree, the PO-witness is a cross-edge in a sibling subtree. The $\forall\mathsf{DR}.A$ fires only on DR-related nodes—the ancestor is PP-related (not DR), so no conflict. The type set T needs a type for the root (containing C_0), one for the PO-witness (containing A), and one for the PP-witness (containing $\neg B$). Cross-edge consistency (CT2) checks that every pair has a non-empty safe set in $\{\mathsf{DR}, \mathsf{PO}\}$—a one-time pairwise check, not an evolving context.

Example 6.2 (Implicit PP via composition forcing)**.** The concept

$$X \;=\; \exists\mathsf{DR}.\exists\mathsf{PO}.C \;\sqcap\; \forall\mathsf{DR}.\neg C \;\sqcap\; \forall\mathsf{PO}.\neg C \;\sqcap\; \forall\mathsf{PP}.X$$

contains no $\exists\mathsf{PP}$ demand, yet it requires an infinite model with forced PP edges.

A node r satisfying X has a DR-neighbor a (with $\exists\mathsf{PO}.C$) and a's PO-neighbor b (with C). By Table 1, the relation between r and b lies in $\mathsf{comp}(\mathsf{DR}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}\}$. But r has $\forall\mathsf{DR}.\neg C$ and $\forall\mathsf{PO}.\neg C$, and $b \in C$—so DR and PO are blocked, leaving *only* PP. The PP edge is **forced** by composition plus universal filtering, with no explicit $\exists\mathsf{PP}$ in the concept.

Since $r \;\mathsf{PP}\; b$, node b is an *ancestor* of r in the cover tree. The recursive $\forall\mathsf{PP}.X$ forces b to also satisfy X, creating its own $\mathsf{DR} \to \mathsf{PO}$ chain and forcing yet another PP-ancestor above it:

$$r \;\mathsf{PP}\; b \;\mathsf{PP}\; b' \;\mathsf{PP}\; b'' \;\mathsf{PP}\; \cdots$$

The model is necessarily infinite. The cover-tree tableau handles this finitely: the type for C-elements can serve as its own PP-ancestor (different domain elements, same abstract state), so a three-element type set $\{\tau_0(C, X),\ \tau_1(\neg C, \exists\mathsf{PO}.C),\ \tau_2(\neg C, \mathrm{root})\}$ represents the infinite model.

This pattern—PP edges arising purely from composition and universal filtering—is a distinctive feature of $\mathcal{ALCI}_{RCC5}$ and illustrates why the complete-graph semantics require careful handling of implicit spatial constraints.

Example 6.3 (The "PO-loop": 2-element SAT via symmetric cycles)**.** Naive reasoning suggests

$$Y \;=\; C \;\sqcap\; \exists\mathsf{PO}.\exists\mathsf{PO}.C \;\sqcap\; \forall\mathsf{PO}.\neg C \;\sqcap\; \forall\mathsf{DR}.\neg C \;\sqcap\; \forall\mathsf{PP}.\neg C \;\sqcap\; \forall\mathsf{PPI}.\neg C$$

should be unsatisfiable: any witness chain $d \;\mathsf{PO}\; a \;\mathsf{PO}\; b$ with $b \in C$ needs $\rho(d, b) \in \mathsf{comp}(\mathsf{PO}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}, \mathsf{EQ}\}$, and the four universals $\forall R.\neg C$ seem to block every non-EQ option (each forces $b \in \neg C$, yet $b \in C$).

But Y is in fact **satisfiable**, via the two-element model $\{d, a\}$ with $C^{\mathcal{I}} = \{d\}$ and $\rho(d, a) = \mathsf{PO}$. Because PO is symmetric ($\mathsf{PO}(d, a) \Rightarrow \mathsf{PO}(a, d)$), the chain $d \;\mathsf{PO}\; a \;\mathsf{PO}\; d$ loops back to the root itself: the "third node" of the $\exists\mathsf{PO}.\exists\mathsf{PO}$ formula *is* d, under strong EQ identity. Two syntactic positions in the nested existential map to the same semantic element.

The universals $\forall R.\neg C$ constrain d's *outgoing* R-neighbours (the sole PO-neighbour $a \in \neg C$ satisfies them), but they do **not** prevent d from *being* a PO-neighbour of a: the witness for a's $\exists\mathsf{PO}.C$ is d itself, and a has no $\forall$PO-constraint forbidding C-neighbours. This asymmetry between "constraints on my neighbours" and "constraints on what I may be a neighbour of" is peculiar to symmetric roles (PO, DR). The cover-tree tableau correctly reports Y as satisfiable.

Example 6.4 ("Clash-out to EQ": genuine UNSAT via strong EQ)**.** The complementary pattern

$$Z \;=\; C \;\sqcap\; \exists\mathsf{PO}.\exists\mathsf{PO}.\neg C \;\sqcap\; \forall\mathsf{PO}.C \;\sqcap\; \forall\mathsf{DR}.C \;\sqcap\; \forall\mathsf{PP}.C \;\sqcap\; \forall\mathsf{PPI}.C$$

is unsatisfiable, and the reason is instructive. The chain d PO a PO c forces $a \in C$ (via $\forall\mathsf{PO}.C$ at d) and $c \in \neg C$ (via $\exists\mathsf{PO}.\neg C$ at a). The relation $\rho(d, c)$ must lie in $\mathsf{comp}(\mathsf{PO}, \mathsf{PO}) = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}, \mathsf{EQ}\}$. The four universals $\forall R.C$ at d (for $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$) each require $c \in C$, contradicting $c \in \neg C$, hence *all four non-*EQ *options are blocked.* Only EQ remains.

Under **strong** EQ **semantics** (EQ = identity), $\rho(d, c) = \mathsf{EQ}$ forces $d = c$, yielding $d \in C \wedge d \in \neg C$ and a clash. The PO-loop escape of Example 6.3 does not apply here: the endpoint c requires $\neg C$ while d has C, so $d \neq c$ is forced semantically, and the universals then close every non-EQ composition entry.

This example illustrates why *strong* EQ semantics is essential for our undecidability- and decidability-proof machinery. Under weak EQ (equivalence-class) semantics, collapsing d and c would not immediately clash, and one would need additional machinery (equality propagation, congruence-closure) to derive the same conclusion. The strong reading is also the one built into the composition tables used in Renz & Nebel's patchwork theorem.

Remark 6.5 (Reasoners and tools: a cross-validation suite)**.** This paper references three Python tools used in the companion implementations; before the incompleteness below we fix their roles. The **cover-tree tableau** (`cover_tree_tableau.py`, Section 6) is the main candidate decision procedure—the only tool whose soundness and completeness are claimed for $\mathcal{ALCI}_{RCC5}$ satisfiability, resting on the patchwork property of RCC5 and the companion completeness-extraction argument. The **quasimodel reasoner** [38] (`alcircc5_reasoner.py`, with cycle-aware wrapper `alcircc5_reasoner_cyclic.py`) is a constructive bottom-up quasimodel search based on disjunctive path-consistency; it is used *only* as a cross-validation oracle for the cover-tree tableau, not as a step of any decidability argument. The **independent model verifier** (`model_verifier.py`) grounds SAT answers in concrete finite RCC5 models, separately checking inverse symmetry, composition consistency on all triples, type-safety, existential witnessing, and recursive concept truth. The three tools together form a cross-validation suite around the cover-tree tableau: cover-tree produces the candidate verdict, the quasimodel reasoner offers an independent opinion, and the model verifier anchors SAT answers in explicit models.

Remark 6.6 (Implementation-level bugs found and fixed in April 2026)**.** Three implementation-level defects were identified in April 2026 and fixed in the cover-tree tableau and the baseline quasimodel reasoner: (i) the baseline quasimodel reasoner was incomplete on cyclic-via-symmetric-role SAT concepts such as the PO-loop of Example 6.3 (closed by an opt-in cycle-close clause in `alcircc5_reasoner_cyclic.py`); (ii) the shared `compute_safe` helper failed to propagate $\forall\mathsf{PP}/\mathsf{PPI}.D$ universals as a whole into successor types (fixed by the standard transitive-role trick); (iii) the cover-tree's pairwise tree-cross check missed twelve UNSAT patterns of the form $\exists R_1.\exists R_2.A \sqcap \bigwedge_{R \in \mathsf{comp}(R_1,R_2)} \forall R.\neg A$ when intermediate types carry a single demand (fixed by porting the quasimodel reasoner's role-path compatibility fixpoint as Phase 5 / (CT5)). The decidability argument is unaffected by any of these fixes: all three were soundness gaps in implementations of the candidate decision procedure, not the split-forest / completeness-extraction argument that underpins the theorem. The 911-concept cross-validation continues to match $911/911$ before and after every fix; full technical detail (root cause, algebraic justification, regression tests) is in the cover-tree implementation paper [29].

6.3. What makes this "non-standard"

The cover-tree tableau differs from a standard DL tableau calculus in several fundamental ways:

1. **No completion graph.** Standard tableaux build a completion graph node by node, applying expansion rules until saturation or clash. The cover-tree tableau searches over *sets of types* (abstract states), not concrete graph structures. It is closer to a type-elimination or automata-theoretic approach.

2. **No explicit blocking rule.** Standard tableaux have an explicit blocking condition that compares nodes during expansion. The cover-tree tableau achieves finiteness *structurally*: the search space is the powerset of $\mathsf{Tp}(C_0)$, which is already finite.

3. **Global side-checker.** The cross-edge constraints (CT2–CT4) are checked globally over the entire type set, not locally during node expansion. This is what avoids the blocking dilemma: cross-edge consistency is a property of the type set as a whole, not of individual nodes.

4. **Existential demands go to ALL existing types.** In a standard tableau, $\exists R.D$ generates one new R-neighbor. In the cover-tree tableau, the witness can be *any* type already in T—including the root type itself. This reflects the complete-graph semantics: every domain element is related to every other.

6.4. Soundness and completeness

The **primary decidability statement** (as of 29 May 2026) is the GPT-5.5 *split-forest* + *parity-tree-automata proof* [27] (21 pages): the split-forest *semantic normal form*—witness thinning, generated containment covers, split copies for incomparable superparts, side-interface mosaics, and typed-EQ quotient—together with a reduction of the finite search problem to *non-emptiness of a parity tree automaton* over the finite alphabet of split-forest profiles. The automaton keeps the split-forest model idea (in the proof's own words, "the automaton does not replace the split forest; it consumes it") and is used only for the parts most brittle in a hand certificate: regular infinite branches, simultaneous eventuality fulfilment, recursive side contexts, and profile repetition without semantic equality. Parity-tree-automaton non-emptiness is decidable by construction, so the finite-checkability and eventuality obligations come for free. This proof was promoted as a strategic move, not a proof upgrade, and was *then cold-reviewed* in turn (the 5th cold review): it was found to share the no-automata route's finite-representation keystone gap and was repaired (Theorem B, finite abstraction adequacy, being the load-bearing keystone). That keystone was *then cold-reviewed in turn* (the 6th cold review, by a fresh Opus 4.8 of a different model lineage from the proof's author) and found unproved—its interface-transformer coherence was asserted, not proved—prompting GPT-5.5's round-12 repair, which deletes the transformer layer and rests global RCC5 consistency on the *patchwork property* of RCC5, decided by a two-way alternating parity automaton. Round-12 has not itself been cold-reviewed. The overall status remains *strongly supported, not certified.*

The *no-automata* route—which encoded those same brittle parts by a finite certificate with request-closed blocking cycles—was the project's primary statement through round 10 and is now its historical predecessor (§7). Its most developed manuscript is the round-9 forced-verticalization proof [23] with companion [24]; a cold review of round-9 then found a dual descendant-tower defect and two foundational gaps (finite checkability; cross-label determinacy), and round-10's repair reduced them to *automaton-shaped* lemmas—which is exactly why the automaton was promoted. The remainder of this subsection develops the shared split-forest normal form and the (now-historical) no-automata certificate machinery, since the automata proof consumes the former and the latter records what the automaton replaces.

The no-automata thread's final state (D-1 and the round-9 repair, historical). The completeness gap (D-1) that a cold Claude Opus 4.8 review found in round-8 was closed at the manuscript level in round-9. The gap was this: round-8's cross side/tree exhaustion lemma closes the side/tree gap only on the *finite, depth-bounded* boundary $B_k(u)$, whereas a DR/PO side witness can be RCC5-composition forced to be a proper part of an *infinite* pumped containment tower. The concrete satisfiable witness is

$C_0' = (\exists\mathsf{PP}.G) \sqcap (\exists\mathsf{PO}.\neg X)$ with $G = (\forall\mathsf{PO}.X) \sqcap (\exists\mathsf{PP}.G)$: the tower $a_1\mathsf{PP}a_2\mathsf{PP}\cdots$ each carries $\forall\mathsf{PO}.X$, and the root's PO-witness $w \in \neg X$ is forced ($\mathsf{comp}(\mathsf{PPI},\mathsf{PO}) = \{\mathsf{PO},\mathsf{PPI}\}$ with PO killed by $\forall\mathsf{PO}.X$, then $\mathsf{comp}(\mathsf{PPI},\mathsf{PPI}) = \{\mathsf{PPI}\}$) to be a proper part of *every* tower node, a label the bounded saturated side context cannot carry (machine-checked). In round-9 GPT-5.5 ratified the defect and integrated the cold reviewer's proposed repair, *composition-forced verticalization*: a side witness whose ancestor trace has a PPI-tail is spliced into the cover relation $E_\uparrow$ at a bounded threshold, so equality-aware vertical reachability carries the forced label up the tail and the u–w edge stays the inherited residual PO pair. Three lemmas support the repair — monotone ancestor traces ($\mathsf{DR}^*\mathsf{PO}^*\mathsf{PPI}^*$, PPI absorbing), a bounded threshold after regular replacement, and a global residual frontier in which every remaining pair is DR or PO. We state the result honestly as **strongly supported but not, as written, mechanically certified**: the soundness direction is sound, the architecture is sound, the D-1 gap is repaired at the manuscript level—though round-9 was then itself cold-reviewed (§7.10), which found a dual defect and two foundational gaps and led to round-10 and the switch to the automata route as the primary statement. GPT-5.5 names the bounded-threshold replacement lemma, the typed-equality quotient lemma, and the support-closed mosaic patchwork theorem as the parts most worth future formalization. The remainder of this subsection states the construction at the level the manuscript develops it.

Soundness (valid certificate $\Rightarrow$ strong abstract RCC5 model satisfying C_0). A finite split-forest certificate $\mathcal{Q}$ is unfolded to an occurrence structure $\mathcal{U}(\mathcal{Q})$ in which repeated cycle profiles generate fresh concrete occurrences. The quotient $\mathcal{U}(\mathcal{Q})/\approx$ by the explicit equality-port congruence is a strong abstract RCC5 frame: strong-EQ holds because the quotient identifies exactly $\approx$-classes; inverse symmetry and RCC5 composition hold because the round-7 label function and triple-composition clause (V6) are verified on the finite occurrence-sensitive certificate. A truth lemma over $\mathsf{cl}(C_0)$ shows that $\mathcal{I}_\mathcal{Q} \models C_0$ at the initial occurrence. The full chain rests on GPT-5.4's v1 sibling-interface paper [10] (Thms 1.17–1.19) with the v2 delta [11], plus the round-7 architectural specifics.

Completeness (satisfiable C_0 $\Rightarrow$ valid finite certificate) — *the direction that carried the D-1 gap, repaired in round-9.* A satisfying interpretation is thinned to a witness-generated model via the witness-thinning lemma; the generated containment split-cover construction supplies a weak split presentation with explicit equality mates and side mosaics. A finite-index blocking lemma (without automata) regularises every infinite vertical ray to a finite prefix plus a request-closed cycle word. The certificate profiles are the resulting complete boundary profiles; pair shapes and triple shapes are extracted from the occurrence-sensitive incidences of the regular presentation, not merely from abstract position symbols. Round-8 filed every DR/PO witness in a bounded side context, which cannot record the labels forced by composition with an unbounded ancestor tower (D-1); round-9 tests each selected side witness for composition-forced verticalization *before* ordinary side-context attachment, splicing those with a PPI-tail into the vertical backbone at a bounded threshold so the unbounded tail is carried by equality-aware reachability (§7).

The round-7 architectural move. The central ingredient of round 7 is that finite labels live on *occurrence-sensitive pair shapes*

$$\alpha(u,v) \;=\; \big(s_u, p_u, s_v, p_v, \iota(u,v)\big),$$

where $\iota(u,v) \in \{\mathsf{self},\mathsf{eq},\mathsf{up},\mathsf{down},\mathsf{side}_R,\mathsf{front}_R\}$ is an *incidence tag* that distinguishes literal occurrence identity ($u = v$, tag self) from explicit equality-port pairs (tag eq), vertical successor / predecessor pairs (tags up, down), side-mosaic pairs, and residual sibling-frontier pairs. The label function is

$$\ell_\mathcal{Q} : \mathsf{Pair}_\mathcal{Q} \to \mathsf{B}, \qquad \ell_\mathcal{Q}(\mathsf{self}) = \ell_\mathcal{Q}(\mathsf{eq}) = \mathsf{EQ},\;\; \ell_\mathcal{Q}(\mathsf{up}) = \mathsf{PP},\;\; \ell_\mathcal{Q}(\mathsf{down}) = \mathsf{PPI},\;\; \ell_\mathcal{Q}(\mathsf{side}_R) = \ell_\mathcal{Q}(\mathsf{front}_R) = R.$$

Distinct laps of a blocking cycle have $\iota \neq \mathsf{self}$ (and, absent an explicit equality-port link, $\iota \neq \mathsf{eq}$), so $\ell_\mathcal{Q}(\alpha(u_i,u_j)) \neq \mathsf{EQ}$ for any distinct laps: the canonical one-state certificate for $\exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top$

is realised with $\iota = \mathsf{up}$ and label PP between successive laps, as intended. Triple composition is verified on a finite triple-shape set $\mathsf{Tri}_{\mathcal{Q}}$ via

$$\ell_{\mathcal{Q}}(\alpha_{13}) \in \mathsf{comp}\big(\ell_{\mathcal{Q}}(\alpha_{12}), \ell_{\mathcal{Q}}(\alpha_{23})\big) \qquad \text{for all } (\alpha_{12}, \alpha_{23}, \alpha_{13}) \in \mathsf{Tri}_{\mathcal{Q}},$$

the round-7 (V6) clause.

Twelve validity conditions, finite enumeration. All of round 7 is twelve finite-checkable validity conditions (V1)–(V12) on a finite occurrence-sensitive certificate. The certificate space is bounded by a coarse $B(C_0) \le 2^{2^{p(n)}}$ for a polynomial p, sufficient for decidability. Three structural ingredients carry the proof:

(1) **Occurrence-sensitive equality.** Equality is generated only by explicit equality-port chains instantiated inside a context, never by reuse of an abstract position symbol in a blocking cycle.

(2) **Split copies for multiple selected proper superparts**. Where an element has incomparable selected PP-superparts, the presentation carries one equality-mate occurrence per selected superpart.

(3) **Request-closed cycles** in place of ω-acceptance. Transitive PP/PPI eventualities are discharged by finite-graph reachability plus a request-closure condition: no parity or Büchi automaton.

Computational verification of the combinatorial claims. Three self-contained pure-stdlib scripts exercise the central round-7 stress concepts: WP7 (comparable side witnesses, $\exists\mathsf{DR}.(A \sqcap \exists\mathsf{PP}.B) \sqcap \exists\mathsf{DR}.B$), WP8 (blocking-not-equality: a finite truncation of the canonical one-state $C_{\uparrow}$ chain verifies that the round-7 incidence-tagged labelling rule keeps distinct laps at PP/PPI rather than EQ, and explicitly demonstrates the discarded round-6 collapsed rule), and WP9 (split copies for $\exists\mathsf{PP}.A \sqcap \exists\mathsf{PP}.B$ with incomparable proper superparts). All three pass. Six earlier work packages (WP1–WP6), written against the round-2 31-page repaired proof [16], also all PASS; their detailed description is in §7 below.

A note on scope. The soundness and completeness statements are theorems about the *split-forest semantics*, not about the implementation in §6. The cover-tree tableau is an algorithmic *realization* of this theory; a formal derivation showing that CT1–CT5 exactly capture the round-7 (V1)–(V12) is a separate claim, empirically supported but not yet written. The companion implementation paper [29] discusses this theory-vs-implementation split in detail.

6.5. Computational evidence

The algorithm is implemented in approximately 350 lines of Python [29, 37]. Cross-validation against an independent quasimodel-based reasoner [38, 39] on **911 test concepts** (50 known SAT, 13 known UNSAT, 36 adversarial, 512 systematic triples, 200 random depth-2, 100 random depth-3) yields **zero mismatches** [42, 43, 44]. A decomposition test [41] confirms that **775/775 SAT concepts (100%)** have finite cover-tree models, with 12.2 M total models enumerated and **zero genuine counterexamples**. Models produced by the tableau are independently replayed by a standalone verifier [40].

Additionally, a GIS taxonomy computation [45] applies the cover-tree tableau to a realistic $\mathcal{ALCI}_{RCC5}$ ontology from Wessel's original report [8]: 18 spatio-thematic concepts (countries, cities, rivers, lakes) connected by RCC5 spatial relations. The tableau reproduces **all 21/21 expected subsumptions**, including non-trivial inferences like Hamburg $\sqsubseteq$ GermanCity that require RCC5 composition reasoning through spatial role chains. Figure 2 shows the original computed taxonomy from Wessel's report [8].

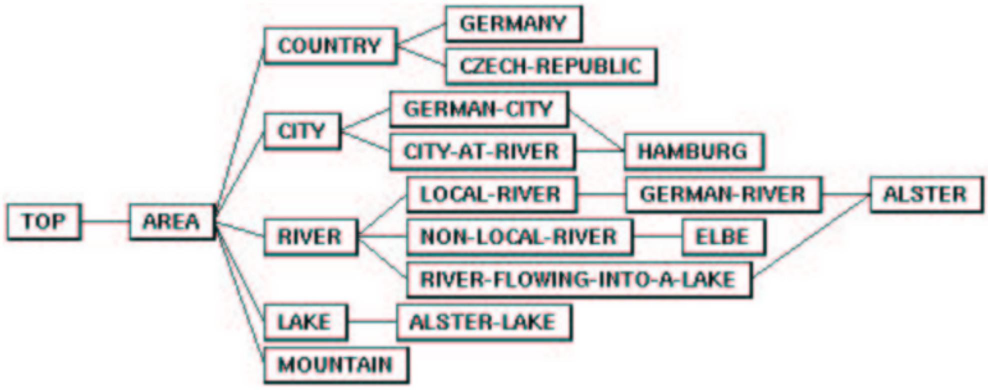

Figure 2: Computed taxonomy of the GIS example TBox from Wessel [8], Figure 6. The cover-tree tableau reproduces all 21 subsumption edges, including non-trivial spatial inferences such as Hamburg $\sqsubseteq$ GermanCity (requiring RCC5 composition propagation through the PO-chain Hamburg–Alster–AlsterLake).

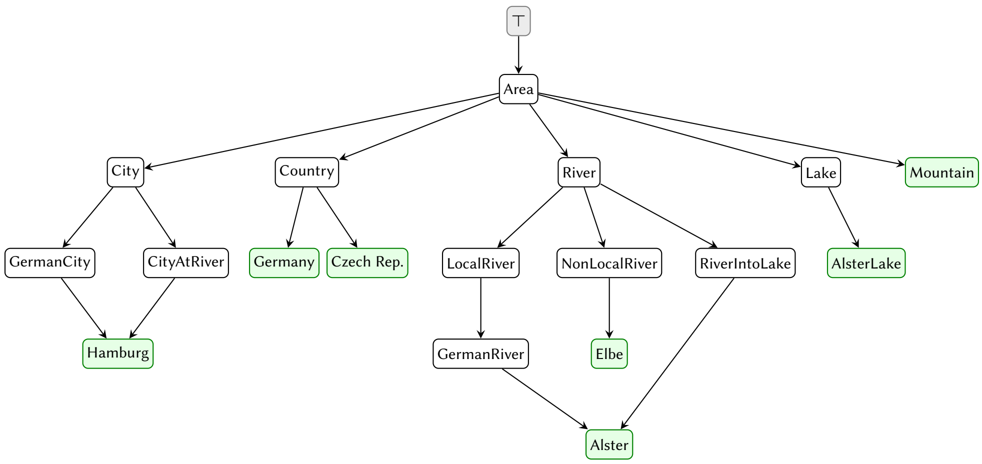

Figure 3: Taxonomy of the GIS example TBox, computed by the cover-tree tableau [45]. Green nodes are leaf concepts (most specific). Note the DAG structure: *Alster* has two parents (GermanRiver and RiverIntoLake) and *Hamburg* has two parents (GermanCity and CityAtRiver). All 21 subsumption edges match Figure 2.

7. History of Attempts (April–May 2026)

The decidability argument summarised in §6.4 reached its current form through seven rounds of adversarial review between Claude (Anthropic) and GPT-5.5 (OpenAI) in April–May 2026, prompted by Michael Wessel. This section gives a compact abstracted narrative of the evolution. The full prose of each round, the original superseded manuscripts, and the per-round repair list are preserved in `OUTDATED.md` [47].

7.1. The starting point: cover-tree tableau + v1 completeness extraction (April 2026)

The cover-tree tableau [12, 29] and its companion split-forest semantics [10] were established in April 2026 by GPT-5.4 Pro and Claude Opus 4.7, prompted by Wessel. The split-forest paper proves the soundness direction in full detail (Thms 1.17–1.19); the converse—extracting a finite quotient from a satisfying model—was given only as a proof sketch. Claude wrote a first attempt at the missing completeness extraction (*v1* [14], April 2026). In parallel, the cover-tree implementation underwent two rounds of adversarial review by Claude Opus 4.7 (April 17–18, 2026): twelve implementation-level counterexamples were identified across the 4×4 grid of (R_1, R_2) three-type chains and the PP/PPI-transitive universal pattern, and all were fixed in the shared `compute_safe` helper and the new Phase 5 / (CT5) role-path compatibility check (Remark 6.6 above). After these fixes the 911-concept cross-validation against the cycle-aware quasimodel reasoner stabilized at $911/911$ matches with zero mismatches.

7.2. v1 superseded, v2.1, and the round-2 31-page repaired proof (May 6–12, 2026)

GPT-5.5 reviewed v1 and identified two structural defects: the Hasse construction for the cover-tree broke on infinite PP-chains (canonical example $C_\uparrow \equiv \exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.\exists\mathsf{PP}.\top$), and the $C5$ triple-coherence proof conflated witnesses across pairs. Claude responded with *v2* [13] (May 6) and *v2.1* (May 6). A second-round review of v2.1 then identified eight structural gaps, three load-bearing: profile-port equality contradicted vertical separation on self-loops via the $M1{+}M7$ reflexivity chain; the mate construction forced all mates to share a single semantic parent (refuted by the two-mate stress concept $C_{AB} \equiv A \sqcap B \sqcap \exists\mathsf{PP}.A \sqcap \exists\mathsf{PP}.B \sqcap \forall\mathsf{PP}.(A \to \neg B) \sqcap \forall\mathsf{PP}.(B \to \neg A)$); and the rootless orientation was incompatible with the depth-from-u_* projection. GPT-5.5 produced a *repaired proof* [16] (May 12, 16 pages) fixing all three defects via occurrence-level equality, mates with different parents, and request-closed cycles in place of ω-acceptance. v2.1 was marked superseded; the repaired proof was adopted as the current best statement. Claude's contribution shifted to verification: six Python work packages (WP1–WP6) cross-checking the repaired proof's finite combinatorial content, all PASS.

7.3. Rounds 3 and 4: structural layers added (May 13–21, 2026)

Rounds 3 and 4 of GPT-5.5 review added four further structural layers to the repaired proof, growing it from 16 to 31 pages: (i) a verticalization discipline (mosaic axiom (M6)) stratifying each context into a vertical stratum and a sibling frontier; (ii) port-indexed mate clusters $\mathrm{Mate}_Q(C, s, p)$ attaching mate equivalences to contexts and ports rather than to bare positions; (iii) an explicit abstract triple graph T_Q with cardinality bounds; (iv) a side interface $\mathrm{Side}(u)$ with polynomial width bound $m(n) = O(n^3)$. The certificate-space bound $B(C_0) = 2^{2^{p(n)}}$ became *grammar-first*, derived from these component bounds.

7.4. Round 5: (M6′) recursive verticalization (May 22–23, 2026)

The round-5 review (GPT-5.5, May 22) identified a load-bearing mathematical defect: the (M6) verticalization discipline was too strong. It forbade non-vertical PP/PPI relations between sibling-frontier mosaic positions, but such relations occur in satisfiable witness-generated models—the round-5 stress concept $\exists\mathsf{DR}.(A \sqcap \exists\mathsf{PP}.B) \sqcap \exists\mathsf{DR}.B$ is SAT in abstract RCC5 yet rejected by (M6). Eight repairs were recommended. Claude drafted two follow-up papers (the (M6′) recursive-verticalization delta and the items-4-to-8 delta) that closed all eight items at the paper level, then consolidated them into a 36-page round-5 v2 paper. Side-witness positions were allowed to open nested local vertical strata, with the round-2 (M6) recovered as the depth-1 case.

7.5. Round 6: internal-coherence pass (May 26, 2026)

The round-6 review of the consolidated v2 (GPT-5.5, May 26) found six internal-consistency gaps: the old side-interface lemma still asserted flat $\{\mathsf{DR}, \mathsf{PO}\}$ sibling frontiers; the weakened per-position support closure (SC4′) was inconsistent with the patchwork proof's per-triple appeal; overlap universal-propagation was under-proved; the lasso construction used the complete profile rather than the extended profile; the side-width bound and validity-clause numbering were inconsistent; and the verification script for the round-5 stress concept was not self-contained. Claude drafted a 39-page round-6 internal-coherence pass closing all six gaps and adding a self-contained pure-stdlib verification script (`wp7_selfcontained_side_witness.py`).

7.6. Round 7: the abstract-position-symbol collapse (May 27–28, 2026)

The round-7 review of the round-6 paper (GPT-5.5, May 27) identified one central architectural defect that could not be locally patched. Round-6 had defined the concrete pair label by $\mathrm{lab}(u, v) := L_{\mathcal{Q}}(q(u), q(v))$, where $q : \Omega(\mathcal{U}(\mathcal{Q})) \to \mathsf{Pos}_{\mathcal{Q}}$ sends each occurrence to an abstract position symbol and $L_{\mathcal{Q}}$ is the abstract pair-label function with standard reflexivity $L_{\mathcal{Q}}(\pi, \pi) = \mathsf{EQ}$. In a finite regular certificate the abstract position alphabet is finite while the unfolding is infinite, so blocking deliberately reuses the same symbol on fresh occurrences. For the canonical one-state upward cycle, distinct laps u_i, u_j share the same symbol π and the round-6 rule gives $\mathrm{lab}(u_i, u_j) = L_{\mathcal{Q}}(\pi, \pi) = \mathsf{EQ}$, even for $i \neq j$: the strict PP chain collapses into a PP-self-loop. The defect is intrinsic to indexing labels by position-symbol pairs.

GPT-5.5's round-7 fix [18] replaces the position-symbol pair labelling by labels on *occurrence-sensitive pair shapes* with explicit incidence tags (described in §6.4). Twelve validity clauses (V1)–(V12) replace round-5 / round-6's fifteen. Claude wrote a round-7 audit/response [28] mapping the round-6 (G1)–(G6) and round-7 (D1)–(D5) defect lists to their resolutions in the round-7 architecture, plus three further self-contained verification scripts (WP7–WP9). All of Claude's round-5 / round-6 manuscripts are superseded by round 7 and retained as the historical audit trail in `papers/gpt5.5_round2/` [47].

7.7. Round 8: saturated side contexts and constructed catalogues (May 28, 2026, evening)

A fresh Claude Opus 4.7 session, given the round-7 sketch with no prior project context, identified three architectural defects. First, the round-7 pair-shape enumeration did not exhaustively cover the case of a side witness paired with a vertical-boundary position: a PP or PPI label between an exposed boundary point and a side position could be hidden in the residual DR/PO frontier closure. Second, the residual DR/PO frontier restriction was imposed *before* the side context was saturated, making the restriction unsound while PP/PPI pairs were still pending. Third, the pair and triple catalogues $\mathsf{Pair}_{\mathcal{Q}}$ and $\mathsf{Tri}_{\mathcal{Q}}$ were asserted finite but not explicitly constructed.

GPT-5.5's round-8 repair [19] closes all three by introducing a *saturated side context* $S_k(u)$ generated by five (Sat1)–(Sat5) operations: start with the exposed vertical boundary of the source, add one selected witness for each local $\exists R.D$ with $R \in \{\mathsf{DR}, \mathsf{PO}\}$ and modal depth strictly less than k, recursively add smaller-depth witnesses, assign or inherit a complete RCC5 label to every ordered pair (including side/tree pairs), and close under equality (label EQ triggers an equality-port declaration) and containment (label PP or PPI triggers a local vertical stratum) *before* applying the residual DR/PO restriction. A new *cross side/tree exhaustion lemma* formalises that no PP/PPI side/tree pair is hidden in the residual frontier. $\mathsf{Pair}_{\mathcal{Q}}$ and $\mathsf{Tri}_{\mathcal{Q}}$ are now constructed as the least finite sets containing the explicit families of pairs and triples generated by the saturated side contexts, overlap amalgams, and residual configurations, with every triple required to satisfy the RCC5 composition table. The 15-page round-8 sketch and the 30-page detailed companion [20] are filed under `papers/claude_latest_review_gpt5.5_fix/`. Round-8 was the most developed statement until a cold Opus 4.8 review found the D-1 completeness gap (next); round-9 is its successor.

7.8. Round 8 cold review: an open completeness gap (D-1) and a proposed repair

A further cold review — this time by Claude Opus 4.8, again with no prior project context — examined whether the saturated-side-context repair actually closes the side/tree gap, and found that it does not: it relocates the gap rather than closing it [21].

The cross side/tree exhaustion lemma quantifies only over the *finite, depth-bounded* boundary $B_k(u)$ and the finite saturated side context $S_k(u)$; both are bounded by the side-width $m_*(C_0)$. The lemma is true as stated, but the advertised global conclusion — "no PP/PPI side/tree pair is hidden" — is a statement about the whole unfolded model, in which the vertical ancestor chain is infinite. The genuinely hard case is a DR/PO side witness that RCC5 composition forces to be a proper part of an *infinite* pumped tower. The concrete satisfiable witness is

$$G := (\forall \mathsf{PO}.X) \sqcap (\exists \mathsf{PP}.G), \qquad C_0' := (\exists \mathsf{PP}.G) \sqcap (\exists \mathsf{PO}.\neg X).$$

G generates an infinite ascending tower $a_1 \mathsf{PP} a_2 \mathsf{PP} \cdots$, each $a_m \in \forall \mathsf{PO}.X$; the root's PO-witness $w \in \neg X$ is forced to satisfy $\rho(a_m, w) = \mathsf{PPI}$ for *all* m (base case: $\rho(a_1, w) \in \mathsf{comp}(\mathsf{PPI}, \mathsf{PO}) = \{\mathsf{PO}, \mathsf{PPI}\}$, and $\forall \mathsf{PO}.X$ at a_1 with $w \in \neg X$ kills the PO option; induction: $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$). Only finitely many tower nodes fit in $S_k(u)$; for the unbounded tail the constructed pair-shape catalogue offers no admissible shape ($\mathsf{side}_{\mathsf{PPI}}$ needs co-residence in one bounded context; down needs a cover-edge path the construction never creates for a side witness; $\mathsf{front}_{\mathsf{DR}}$ is composition-illegal; $\mathsf{front}_{\mathsf{PO}}$ violates the $\forall \mathsf{PO}.X$ universal-safety check). So the round-8 completeness construction produces an invalid certificate for the satisfiable C_0' — a genuine gap ("D-1"), of medium–high severity. The composition arithmetic is machine-checked.

The decidability *theorem* most likely survives: placing w as a true vertical descendant of the tower (a selected $E_\uparrow$-child of a_1, with the u–w edge demoted to a residual PO frontier) appears to yield a valid certificate, and the decision procedure accepts whenever *some* valid certificate exists. The accompanying repair note [22] turns this observation into a construction — *composition-forced verticalization*: detect when composition forces a DR/PO witness to be a proper part of an ancestor, and splice an $E_\uparrow$ cover edge at the (provably bounded) threshold height where it first becomes a proper part, so equality-aware reachability carries the correct PP/PPI label up the entire tail. Three facts make the repair clean, the first machine-checked: (i) the forced label up an ascending chain is monotone in $\mathsf{DR} \succ \mathsf{PO} \succ \mathsf{PPI}$ and PPI-absorbing, hence eventually constant at a threshold; (ii) the threshold is bounded by the number of boundary descriptors, so finiteness is preserved; (iii) the construction *presents a fixed consistent model*, so RCC5 composition and universal safety are inherited rather than re-derived. The cold reviewer recorded this as a candidate with open bookkeeping obligations; GPT-5.5 then took it up in round-9 (next).

7.9. Round 9: composition-forced verticalization closes D-1 (May 29, 2026)

GPT-5.5 ratified the D-1 defect and integrated the cold reviewer's repair into a round-9 proof [23] (16-page sketch) with detailed companion [24] (33 pages), filed under `papers/gpt5.5_round3/`. A 5-page formal response memo [25] records that GPT-5.5 re-ran the three Opus 4.8 verification scripts (all pass), confirmed the composition facts $\mathsf{comp}(\mathsf{PPI}, \mathsf{PO}) = \{\mathsf{PO}, \mathsf{PPI}\}$ and $\mathsf{comp}(\mathsf{PPI}, \mathsf{PPI}) = \{\mathsf{PPI}\}$ that drive the phenomenon, and agreed that "this is a genuine completeness defect, not merely a presentational ambiguity."

Round-9 distinguishes two cases for a selected DR/PO side witness w of a source u. If its relation to the ancestor tower remains DR/PO, w stays an ordinary side witness and deep ancestor–witness pairs are residual DR/PO pairs. If RCC5 composition forces w to be a proper part of all sufficiently high tower nodes, the proof inserts a generated cover edge $w' \, E_\uparrow \, a^*$, where a^* is the first retained representative of the eventual PPI-tail and w' is w or a split equality mate; every later pair (a_m, w') is then presented by equality-aware vertical reachability. This is a presentation change, not a semantic change: the splice is added only when the inherited satisfying model already has $w' \mathsf{PP} a^*$, and the u–w relation stays the inherited DR/PO residual pair.

Three lemmas carry the repair. *Monotone ancestor traces*: for a DR/PO witness w of u and a tower $u < a_1 < a_2 < \cdots$, the trace $r_m = L(a_m, w)$ has shape $\mathsf{DR}^*\mathsf{PO}^*\mathsf{PPI}^*$ (EQ, PP impossible; PPI absorbing because $\mathsf{comp}(\mathsf{PPI}, \cdot)$ never increases rank). *Bounded threshold after regular replacement*: augmenting the tower descriptor with the three-valued trace status leaves finitely many descriptors, so each of the at most three constant regimes needs only boundedly many representatives and the first retained PPI-point sits at a computable height $N_{\mathrm{thr}}(C_0)$. *Global residual frontier*: after equality ports, finite containment saturation, and forced verticalization, every remaining residual-frontier pair has label DR or PO. The split-cover and extraction proofs are updated to apply the forced-verticalization test before ordinary side-context attachment.

The repair preserves the no-automata split-forest architecture (no Büchi/parity automata); the only extra finite information is the three-valued ancestor-trace status, absorbed into the existing finite-index replacement argument. On the computational side, the round-9 construction is exercised end-to-end by a certificate-emission test on C_0' and its UNSAT sibling, generalised to a parameterised forced-tower family and cross-checked against a by-construction oracle and the cover-tree tableau (§7.12); this closes the round-8 review's recommendation to add C_0' as a constructive emission test. GPT-5.5 explicitly characterises round-9 as "a serious proof manuscript rather than a mechanically certified theorem," naming the bounded-threshold replacement lemma, the typed-equality quotient lemma, and the support-closed mosaic patchwork theorem as the parts most worth future formalization. When written, round-9 was *strongly supported but not certified* and the most developed statement of the no-automata thread. It was then cold-reviewed in turn—which found a dual descendant-tower defect and two foundational gaps—and superseded within the day, as the next subsection records.

7.10. Round 9 cold review, round 10, and the switch to the automata route (May 29, 2026)

A fresh GPT-5.5 session, given the round-9 manuscripts with no project context, reviewed round-9 and found it was not yet a proof. The composition table was independently re-verified correct. Three *critical* defects:

- **One-sided verticalization.** Round-9 handled a DR/PO witness forced into an infinite *ancestor* PPI-tower but missed the dual case — a PO witness forced to be a proper *superpart* of an infinite *descendant* PP-tower. Concrete witness $C = (\exists\mathsf{PO}.A) \sqcap (\exists\mathsf{PPI}.\top) \sqcap \forall\mathsf{PPI}.(\exists\mathsf{PPI}.\top \sqcap \forall\mathsf{DR}.\neg A \sqcap \forall\mathsf{PO}.\neg A)$: each descendant d_i is forced ($\mathsf{comp}(\mathsf{PP},\mathsf{PO}) = \{\mathsf{DR},\mathsf{PO},\mathsf{PP}\}$, DR/PO killed by the universals, $\mathsf{comp}(\mathsf{PP},\mathsf{PP}) = \{\mathsf{PP}\}$ absorbing) to satisfy $d_i \mathsf{PP} w$ (machine-checked).
- **Finite checkability never proved.** The validity clauses (V6)/(V7) were *asserted* to be finite syntactic checks but quantify over the infinite unfolding; the finite pair/triple catalogues were never shown to exhaust it.
- **Boundary non-determinacy.** A boundary descriptor does not determine the mixed outside/internal labels: $i\mathsf{PP}b,\ o\mathsf{PO}b \Rightarrow L(o,i) \in \{\mathsf{DR},\mathsf{PO},\mathsf{PPI}\}$.

plus four majors, including a *false* simple-cycle bound (a request-closed cycle may need a closed walk that revisits descriptors).

Round 10 accepts all findings and repairs them: a dual descendant splice (symmetric to round 9), occurrence-local requests, ranked side-width, global-quotient-before-cycle-selection, a corrected $(M+1)N$ closed-walk bound, and — for the two foundational criticals — a *finite product-state exhaustion* construction and *complete interface descriptors*. A close reading finds that the entire decidability burden then funnels into two named lemmas, Finite Product Exhaustion and Complete-Interface Replacement, both established by case analysis plus finiteness of a descriptor alphabet, and both *automaton-shaped*: product-state exhaustion is product-automaton reachability, and the complete interface descriptor is a finite interface abstraction.

This is the convergence point. Over rounds 2–10 the no-automata route repeatedly rediscovered automaton-shaped obligations (request-closed cycles for ω-acceptance; product-state exhaustion for finite checking) that it could not discharge. The decision (29 May 2026) is therefore to promote the split-forest + parity-tree-automata proof [27] (§6.4) to the primary statement: it keeps the entire split-forest normal form and lets a parity tree automaton — whose non-emptiness is decidable by construction — handle the brittle half directly. The promotion was strategic, not a proof upgrade; the automata proof was then cold-reviewed in turn (the 5th cold review), found to share the finite-representation keystone gap, and repaired; its finite-abstraction keystone (Theorem B) was then cold-reviewed in turn (the 6th cold review) and found unproved, prompting GPT-5.5's round-12 patchwork-bag repair (global RCC5 consistency via the patchwork property, decided by a two-way parity automaton), which is now the canonical statement and has not itself been cold-reviewed.

7.11. What survives, and why

The split-forest architecture itself—witness thinning, generated containment covers, weak-EQ split copies for multiple selected PP-superparts, sibling interfaces recorded by finite support-closed mosaics with $\{\mathsf{DR}, \mathsf{PO}\}$-only open cross-relations, typed equality-port congruence quotient to recover strong-EQ semantics, and request-closed cycles—is present in every round from v1 through round 7. What changed was *how labels are indexed on this architecture.* The v1, v2, v2.1, and round-2 versions indexed labels by the rank-d quotient and abstract position symbols; the round-7 architecture indexes them by occurrence-sensitive pair shapes. The whiteboard sketch (Wessel), the DR-rigidity insight, the EQ-splitting and typed-EQ-quotient idea, the cover-tree decomposition, GPT-5.4's v1 formalisation, and Claude's cover-tree tableau implementation all remain load-bearing in round 7.

7.12. Computational verification across rounds

In parallel with the proof work, nine Python work packages verify the finite combinatorial content of the proof family. WP1–WP6 verify the round-2 31-page repaired proof [16] and all PASS: brute-force enumeration confirms the RCC5 composition table in all 25 cells (WP1); a validity-condition checker accepts $C_{\uparrow}$ via parent self-loops, accepts C_{AB} via two mates with different parents, and rejects the v2.1-style single-parent C_{AB} attempt (WP2); a small-model SAT/UNSAT oracle exhibits a concrete 3-element model for C_{AB} (WP3); C_{AB} and request-closed cycle tests fold into WP2/WP3 (WP4, WP5); a mosaic enumeration confirms support-closure axioms on small mosaics (WP6). WP7–WP9 verify the round-7 central stress concepts (comparable side witnesses, blocking-not-equality, and split copies for incomparable proper superparts). WP10 drives the D-1 witness C_0' end-to-end through the round-9 forced-verticalization construction—detect the forced ancestor trace, splice at the bounded threshold, build the incidence-tagged certificate, validate the round-9 clauses—emitting a valid certificate for C_0' and rejecting its UNSAT sibling; WP11 generalises this to a parameterised forced-tower family (period-p profiles, many side witnesses, universals over $\{\mathsf{PO}, \mathsf{PPI}, \mathsf{DR}, \mathsf{PP}\}$) and cross-checks the round-9 emit/reject verdict against a by-construction oracle, an independent certificate validator, and the cover-tree tableau (216/216 agreement on the current sweep, no completeness or soundness gap found on that family; the overlap-amalgam case remains future work). WP10/WP11 close the round-8 review's recommendation to add C_0' as a constructive certificate-emission test. All PASS. No counterexample to any combinatorial claim of any round was found.

8. Outlook and Conclusion

8.1. Preliminary conclusion (end of May 2026)

We state the status plainly, as it is provisional. The decidability of $\mathcal{ALCI}_{RCC5}$ *remains open* (and $\mathcal{ALCI}_{RCC8}$ likewise); as of the end of May 2026 the decidability argument is *strongly supported but not certified.* Two layers should be kept apart.

The settled foundation is the split-forest model class—the semantic normal form of Theorem A (witness thinning, generated containment covers, split copies for incomparable superparts, side-interface mosaics, typed-EQ quotient). Every satisfiable concept admits a witness-generated, tree-like split-forest model, and this normal form has been found sound by *every one of the six cold reviews* conducted on the project—it is the one component that has never broken. Everything downstream is built *on top of* it: the finite patchwork-bag abstraction and the two-way alternating parity tree automaton *consume* the split forest rather than replace it.

The open layer is exactly that finite abstraction. The current canonical manuscript (round 12) replaces the earlier transformer-based abstraction—which a cold review found unproved—by a patchwork-bag abstraction whose global RCC5 coherence follows from the patchwork property of RCC5, with infinite eventualities decided by two-way parity-automaton emptiness. It is a complete, self-contained manuscript that *proves* its finite-abstraction adequacy on paper, modulo two standard external theorems (the RCC5 patchwork property; emptiness of two-way alternating parity tree automata). A subsequent self-contained 29-page edition makes this fully explicit, expanding the theorem spine with appendices that discharge every load-bearing step—in particular deriving the global RCC5 labelling by a compactness argument over the finite relation alphabet (given finite patchwork), with no route-dependent transformer, which is precisely the global-coherence step the cold review had faulted in the earlier transformer-based abstraction. What the manuscript has *not* yet received is an independent cold review or a machine-checked certification. "Strongly supported, not certified" should be read in precisely this sense—proved on paper, not independently verified—and not as a claim that any step is known to be missing. *(Update, July 2026: that last clause has since been overtaken—the seventh review identified one concretely unproved step; see §8.2.)*

The trajectory across the rounds is convergent: the repairs have moved from bespoke, hand-built machinery towards standard, citable theorems, so we regard the present state as a natural consolidation point. Settling the question definitively will require either mechanization (Lean/Coq) of the finite-abstraction keystone—axiomatizing the two external theorems and formalizing the normal form and the adequacy argument—or human expert peer review. Until then we offer the work as a discussion piece: a problem open for over twenty years, with the standard undecidability routes shown blocked and a convergent-but-uncertified decidability argument resting on the sound split-forest model class.

8.2. Update (July 2026): the seventh review

The consolidation stance above held for five weeks, until a genuinely stronger reviewer became available: Claude Fable 5 (Anthropic's first Mythos-class model, a capability tier above Opus, and a different model lineage from the round-12 author, GPT-5.5). A materially stronger reviewer is itself new information, so the freeze was broken once for a *seventh review*—a context-loaded ("hot") adversarial audit of the canonical round-12 manuscript, explicitly *not* a cold review, accompanied by three new computational work packages (WP14–WP16); see `papers/fable5_review/` in the repository for the full report and probes. It continued the series' pattern: every review so far has found a defect, and this one did too (seven reviews, seven defects).

The critical finding is that the round-12 truth lemma rests on an unproved *shadow-amalgamation* step: the global RCC5 relation ρ_T is amalgamated by patchwork from the *bag networks only*, yet the lemma assumes the declared relation-vector shadow entries agree with ρ_T. No legality test constrains two shadows against each other, and the implicit lemma—bags plus all declared shadow rows are jointly realizable in one frame—is in fact false for the tests as written: WP15 exhibits a three-bag configuration (declared entries $x_1\,\mathsf{DR}\,x_2$, $x_1\,\mathsf{PP}\,p$, $x_2\,\mathsf{PPI}\,p$; $\mathsf{comp}(\mathsf{PP},\mathsf{PP}) = \{\mathsf{PP}\}$ forces $x_1\,\mathsf{PP}\,x_2$) that passes every listed test and admits no realizing frame; 23 of 64 assignments on the minimal pattern fail. This is the sixth review's transformer-coherence gap recurring one layer up, but strictly smaller: the single-shadow case is provable from the patchwork theorem itself, and no counterexample exists under a *vertical-liveness* repair discipline ($\mathsf{PP}/\mathsf{PPI}/\mathsf{EQ}$ pairs always co-bagged, shadow entries restricted to $\{\mathsf{DR},\mathsf{PO}\}$—the saturation rules extended to the shadow layer; 358/358 random and exhaustive configurations clean) or under an extended-bag discipline (200/200). Two specification loopholes were also found (EQ-valued shadow entries; undefined shadow propagation scope), each independently soundness-fatal as literally written and each a one-line fix: WP16 machine-checks that either loophole makes the mechanism wrongly accept four unsatisfiable forced-composition concepts, while under the repaired discipline it agrees with the cover-tree tableau on all ten referee diagnostics—including, notably, the strong-EQ PO-loop concept that the retired quasimodel reasoner decided wrongly—and on a 148-concept random sweep without disagreement. The $K(C_0)$ finiteness argument also remains an assertion.

What held up under attack is equally important: the RCC5 patchwork theorem *in exactly the round-12 form* was verified exhaustively at small widths (WP14: 63,219 two-bag amalgamations and 300 tree-of-bags patchings executed literally along the manuscript's own induction, zero failures), as did the finite-to-countable compactness passage, strong equality, the running-intersection layer, and Theorem A's normal form. No accepted-but-unsatisfiable *concept* was found. The status label is therefore unchanged—*strongly supported, not certified*—with the keystone sharpened from "patchwork-bag adequacy" generically to a *shadow-amalgamation lemma* (plus the $K(C_0)$ accounting) specifically. The consolidation stance otherwise stands: mechanization or human peer review remain the decisive next steps, with frontier-model audits as the sanctioned exception whenever a genuinely stronger reviewer appears.

Round 13 (July 2026, unreviewed). The seventh reviewer then executed its own repair programme (`papers/fable5_round13/` in the repository): a ten-page delta on round-12 that tightens the abstraction language — shadow entries restricted to the horizontal relations $\{\mathsf{DR}, \mathsf{PO}\}$, with vertical and equality relationships co-bagged (the existing saturation and residual-frontier regime extended to the shadow layer), mandatory shadow propagation with a verticalization fallback, type-propagation of transitive vertical universals, request-source liveness, and a pair-coherence rule — and, under that discipline, *proves* the previously missing shadow-amalgamation lemma from four exhaustively machine-checked composition-table facts plus the patchwork-and-compactness apparatus; the truth lemma then holds with the declared shadow entries literally equal to the amalgamated relation. The residual proof burden moves, explicitly, to the completeness side (a coverage lemma, and the interaction of on-demand verticalization with the effective width bound), which the manuscript designates as the next review target; the effective width bound itself is re-derived from a witness-generated accounting. A further harness (WP17) checks the finite facts exhaustively and the lemma's constructive content on four hundred random tree-shaped configurations, with a functioning negative control. Round 13 is authored by the reviewer, is unreviewed, and inherits the project's presumption of a defect until a genuinely cold review fails to find one; the status label is unchanged: strongly supported, not certified.

Round 14 (July 2026, GPT-5.5; the eighth review). The presumption was correct. Handed a targeted task packet for round 13's self-designated weak points, GPT-5.5 broke the round-13 extraction clause exactly there (the eighth review; eight reviews, eight defects): the satisfiable tower concept $\exists\mathsf{PP}.\top \sqcap \forall\mathsf{PP}.(\exists\mathsf{PP}.\top \sqcap (\forall\mathsf{PP}.\exists\mathsf{PP}.\top \sqcap \forall\mathsf{DR}.A))$ gives every tower level a horizontal universal obligation, and round 13's literal propagation-with-verticalization clause then forces every remote level into the root bag — unbounded width for a fixed concept, machine-checked; a second finding shows mixed PP/PPI paths defeat the orientation assumption of the type-propagation rule. The shadow-amalgamation lemma itself survived re-testing. GPT-5.5's round-14 repair withdraws the horizontal-shadow discipline in favour of the *active-virtual-bag* form of the extended-bag route (the alternative the seventh review had outlined): every universal source is treated as a virtual active occurrence at every bag, with exact — possibly vertical — relation vectors; validity requires every *finite* restriction of an active bag to be a closed complete atomic RCC5 network, with adjacent agreement; patchwork and compactness then realize all declared values verbatim, the coverage lemma becomes trivial, and extraction adds no live ports (the tower is absorbed with two live ports and virtual shadows). Claude reproduced all seven new harnesses and hot-audited the manuscript; the designated cold-review targets are the automaton's pair/triple monitor bookkeeping (self-flagged by the author), the explicitness of the single-shadow closure check, and the inherited width accounting. Two structural observations deserve record: both the seventh and eighth defects were found precisely where the preceding round's "what would invalidate" list pointed, and the two model lineages have now each broken and each repaired the other's round — the review-and-designate discipline is functioning as designed. Status unchanged: strongly supported, not certified; a genuinely cold review of round 14 is the next step.

The ninth review (July 2026, cold): the decision layer is the gap. Round 14 was then reviewed *cold* by a fresh session of the strongest available model, per the project's packet protocol (manuscripts only; all prior reports withheld). The verdict was *gap*, with the most precise delimitation the project has had. On the positive side, the reviewer *reconstructed* — not merely read — the active-virtual-bag semantic layer and found it sound, calling its amalgamation and truth lemmas the strongest versions of these results in the stack's history; strong equality survived its sharpest uniqueness probes in both directions; the composition table and the small-width patchwork instances were re-derived from scratch. On the negative side, the critical finding is structural: a valid active-virtual-bag abstraction is a *pair* — a finitely-labelled tree plus an unbounded relational enrichment that is not carried by any label — so the automaton theorem must decide an existential-*projection* language, a statement the stack never formulates; and the sketched monitor design is *unsound as described*, because independent alternating run copies cannot be forced to guess the off-label data consistently (the reviewer exhibits a machine-checked configuration in which every sketched check passes while no coherent enrichment exists, the required value being pinned to the empty set $\mathsf{comp}(\mathsf{PP},\mathsf{PP}) \cap \mathsf{comp}(\mathsf{PP},\mathsf{DR}) = \varnothing$). Three natural repairs (label annotation, profile quotienting, canonical guessing) are each shown blocked by facts the stack itself establishes. Two further findings concern the width layer: the liveness cost of deferred existential requests is never integrated into the width accounting, and the surviving width lemma's recurrence does not close as written (its witness generation can cascade at fixed rank; the identified fix is a single decreasing measure). The reviewer's structural summary deserves quotation: across the last three rounds "the gap has migrated; it has not closed" — from bounded labels with an unproved existence claim, to restricted rows that broke width, to total semantic data with no decision procedure. The ledger stands at nine reviews, nine defects; for the third time, a cold review found what context-loaded audits had missed. The recommended round-15 alternatives are a bounded-interface or uniformization lemma making the enrichment label-representable, or abandoning the tree-automaton layer for a mosaic/type-elimination decision procedure running directly on the infinitary abstraction, whose model-construction half the review certifies as already in place. Status unchanged: *strongly supported, not certified* — a semantic theory in the best shape it has ever been, and a decision layer that remains unwritten.

Round 15 (July 2026): the decision layer, without automata. Executing the ninth review's second alternative, round 15 removes the automaton layer entirely and replaces it by a *cluster quasimodel* certificate: a finite set of Hintikka types with safe links, a finite catalogue of small closed atomic cluster patterns in which every existential demand is fulfilled within one gluing step (so there are no promissory witnesses and no parity condition at all — the deferral defects of the early rounds are structurally excluded), long-range conditions over a newly isolated *fold lattice* — the closure of the RCC5 atoms under set-composition, shown by exhaustive machine check to contain exactly ten sets, each a vertical singleton or containing a horizontal atom — and a shared core cluster for uniquely realizable types (absorbing the ninth review's shared-witness probe with no liveness cost). Global coherence is constructed once, in the soundness proof, by iterated patchwork over the unfolding together with a canonical horizontal selection on the fold lattice; no run-time device guesses anything, so the ninth review's critical finding does not apply structurally. The width bound is re-proved under a single strictly decreasing measure, resolving the review's third finding, and the sole remaining external input is the RCC5 patchwork property — the two-way parity-automaton emptiness dependency is gone, which also shortens the eventual mechanization path. A new harness checks the fold lattice exhaustively, probes the steering keystone on 250 random glue steps (all realizable; a negative control fails as it must), and cross-validates the acceptance semantics against the cover-tree tableau on all eleven historical diagnostics plus a 200-concept random sweep with no disagreements. Round 15 designates its own weak points (the steering induction, the horizontal normal form for extraction, and one-step saturation at bounded width), is authored by the same lineage that wrote round 13, and is unreviewed: after nine reviews and nine defects it should be presumed to contain the tenth until an adversarial review — next, by the other lineage — fails to find it. Status unchanged: *strongly supported, not certified.*

What survives of the split-forest models in round 15. Since the decision layer has now changed shape three times, it is worth stating precisely how much of the original split-forest model class (GPT-5.4, April 2026) is still load-bearing. The answer is: essentially all of the model theory, and none of the successive decision machinery. Witness thinning is unchanged and remains the first step of the extraction. The saturated side contexts survive *verbatim* — the round-15 cluster patterns are exactly the saturated bags of Theorem A, and the one-step demand condition is the saturation discipline restated.

Split copies for incomparable superparts survive as the gluing interfaces of the pattern catalogue (running intersection playing the role of the old overlap maps). The generated containment forest survives as the *unfolding*: infinite towers are infinite chains of pattern copies, and the old vertical-closure lemma ($\mathsf{comp}(\mathsf{PP},\mathsf{PP}) = \{\mathsf{PP}\}$) survives as the transitive-safety condition together with the singleton case of the fold analysis. Strong equality as exact occurrence identity survives and is extended by the core cluster, which gives semantically unique elements one global occurrence rather than quotienting copies. Finally, the residual-frontier principle of the early rounds — only DR/PO may remain unrepresented — reappears, generalized and machine-checked, as the fold-lattice fact that every unrepresented long-range relation is chain-forced vertical or admits a horizontal value. What has been discarded along the way is exactly the succession of decision devices: the hand-written certificate clauses of rounds 2–10, the interface transformers of round 11, the relation-vector shadows of rounds 12–13, the guessed enrichment and parity automata of rounds 11–14. In this sense the round-15 certificate is the most literal finite image of the split-forest model class the project has produced: it *is* the split forest, folded by pattern isomorphism, with the fold lattice bounding what the folding forgets.

The tenth review (July 2026): round 15 falls at its designated keystone. GPT-5.5's targeted pass broke round 15 exactly where the manuscript pointed: the steering condition of the canonical completion is stated pairwise (each unrepresented pair must admit a safe horizontal value in its fold), and a minimal machine-checked witness — two safe sets $\{\mathsf{PO}\}$ and $\{\mathsf{DR}\}$ over an already-realized PP edge, with $\mathsf{comp}(\mathsf{PP},\mathsf{DR}) = \{\mathsf{DR}\}$ pinning the pair the first set cannot reach — shows the pairwise condition passes while no joint completion exists. The post-mortem sharpened the finding: the witness configuration is semantically unrealizable, so the condition accepts what it should reject, and the repair — making the condition *joint*, requiring a closed completion of the finite domain problem per gluing profile — provably cannot hurt completeness, since extraction from a real model always supplies the joint solution. The review also exposed a spec-versus-test mismatch in the accompanying harness, which had validated a stronger discipline than the manuscript stated; the harness now tests the condition as written and confirms the defect at scale. The ledger stands at ten reviews, ten defects, and the failure genus is the project's single recurring shape — pointwise checks that do not compose jointly — in its fourth incarnation. Two things did not break, again: the split-forest model class, and the round-15 architecture around the failed condition (the fold lattice, one-step clusters, and the core were not reached). The round-16 direction is fixed and small: replace the pairwise steering condition by the joint per-profile amalgamation check. Status unchanged: *strongly supported, not certified.*

Round 16 (July 2026): the joint selection becomes certificate data. The repair of the tenth defect implements the reviewer's own closing recommendation: the certificate now stores, per gluing edge, a *steering function* from finite interface states (type, row over the separator, safety signature) to fresh rows, subject to four finite conditions including a pair condition over the *reachable* pair states — computable from the certificate itself by a least fixed point, because every long-range value in the unfolding is generated by the certified functions. The canonical completion of the soundness proof thereby becomes deterministic: no pointwise choice survives anywhere, each triangle class is discharged by a certified condition, and the tenth review's witness family is provably rejected (machine checks: soundness on all accepted random glue steps, zero false accepts). The residual debt is concentrated in one named and *measured* obligation: the uniformization lemma — extraction must factor the model's per-instance values through states — which at the coarsest granularity fails on 1.4% of random jointly-realizable steps, with two finite repair routes identified; soundness does not depend on it, so the risk direction is over-rejection only. The ledger stands at ten reviews, ten defects; round 16 is reviewer-authored and unreviewed, and should be presumed to contain the eleventh. Status unchanged: *strongly supported, not certified.*

The eleventh review (July 2026, cold): the first no-counterexample verdict. The round-16 stack was cold-reviewed by a fresh session of the other lineage. For the first time in eleven reviews, the referee found *no counterexample and no false lemma*: the verdict is "gap — not certified as written," with the explicit assessments that no algebraic counterexample to the intended theorem was found, that the four-class triangle taxonomy is algebraically adequate under strengthened invariants, that the uniformization delimitation is sound (failure would mean over-rejection, not wrong answers), and that with six named repairs "the intended soundness argument [is] very likely correct." The repairs demanded are formalization obligations rather than corrections: an explicit finite transition system for the reachability fixed point, an ordered-pair transport rule (supplied verbatim by the reviewer), a proved operational-inclusion invariant including the shared-core case, the separator-covers-core statement, explicit pair orientation, and stored witness choices; the width budgets remain the standing deeper item, alongside uniformization. The reviewer independently re-derived the ten-set fold lattice, matching the project's computation exactly. The ledger is now eleven reviews: ten defects with witnesses, and one demand for unwritten formalization — the qualitative shift the choice-free round-16 design was built to force. The status label does not change on one such verdict: *strongly supported, not certified.*

Round 17 (July 2026): the work order implemented. Round 17 supplies everything the eleventh review demanded: the gluing normal form (every pre-existing member of a new pattern copy lies in the separator; separators consist of parent-copy members and core references; the core is a permanent interface at every step), the explicit finite transition system (singleton states with persistent core rows and a total three-case row decomposition; ordered pair states with simultaneous converses, seeded, created at birth, and transported by the reviewer's own rule, underpinned by a no-overwrite lemma), a *proof* of the operational-inclusion invariant — every state and every ordered pair actually occurring in an unfolding lies in the computed reachable sets, with the shared-core cases discharged by core persistence and birth-pattern membership — and a stored witness map making the canonical unfolding a function of the certificate, so the determinism claim is exact. An executable rendering of the system verifies the inclusion invariant on twelve hundred random unfoldings without a violation and confirms that the transport rule is load-bearing (birth-only fixed points miss transported pairs in every catalogue that has them). By the eleventh reviewer's stated criterion, the soundness half of the decision layer is now fully written — explicit system, proved inclusion, no free choices — and, being literally in transition-system form, has become a transcription target for mechanization rather than a research problem. The delta is unreviewed; the completeness side retains its two named open lemmas (the width budgets and uniformization). Status unchanged: *strongly supported, not certified.*

The twelfth review (July 2026): the transport rule has a seam. Framed as an acceptance test against the eleventh review's own work order, the twelfth review rejected round 17 on a precise, finite finding: the written ordered-pair transport rule fires only when both endpoints are updated by the ambient transition, while a pre-existing member of the parent copy outside the child separator receives a birth-adjacent catalogue state instead — so mixed ambient/parent-copy pairs silently escape the fixed point, the pair condition never inspects them, and the round-16 counterexample triangle recurs at exactly that seam (machine-checked, reproduced). Two instructive meta-findings accompany it: the executable harness had implicitly implemented the correct *total* transport, which is why twelve hundred simulations showed nothing — the defect lives in the written rule, not the intended construction — and the harness itself was caught being hash-seed-nondeterministic. The stack also grew stronger where it held: the reviewer failed to break the parent-plus-core normal form and volunteered the running-intersection argument for its extractability, found no overwrite witness, and discharged the ordered-pair item cleanly. The ledger stands at twelve reviews: eleven defects with witnesses, one no-counterexample verdict. The round-18 repair is fully specified by the reviewer (a total endpoint-update operation, transport under it, the re-proved inclusion case analysis, defined path semantics, and a deterministic branching harness). Status unchanged: *strongly supported, not certified.*

Round 18 (July 2026): total transport, and the birth flip. Round 18 implements the twelfth review's repair package in full, with one sharpening that working out the demanded path semantics forces: the total endpoint update has *three* clauses, not two. Besides the review's ambient case (recorded steering rows carried forward) and resident case (catalogue read-off), cross-branch occurrences require a *birth-flip* clause — a younger occurrence's entry to an older separator member is the converse of the steering value the older side received at the younger's birth — which is precisely the least-common-ancestor role change the review had flagged as undefined. A birth-dichotomy lemma (at every step, every pre-existing occurrence is co-patterned or steered; there is no third case) makes the update total and every pair value single-assignment; pairs are transported in lockstep; the inclusion theorem is re-proved by a five-way endpoint-class analysis in which the review's mixed seam becomes an ordinary transport step; and an order-invariance lemma derives the path semantics. The accompanying harness makes the review's methodological lesson structural: states are computed from the written rules alone, never from the realized structure, and cross-checked against ground truth on genuinely branching unfoldings — one hundred and fifty runs without divergence, thousands of mixed and flip pairs exercised, a constructed negative control, and hash-seed-deterministic output; its own development re-derived, empirically, the review's injective-slot-map discipline before it was enforced. The two open completeness lemmas (width budgets, uniformization) are unchanged; the delta is unreviewed, and after twelve reviews the presumption of a further finding applies. Status unchanged: *strongly supported, not certified.*

The thirteenth review (July 2026): the flip vindicated; the prose still short. The thirteenth review rejected round 18 while confirming its central claim: the birth-flip clause is necessary — the previous review's own two-case update was incomplete, exactly as round 18 had argued — and correct as far as it goes. What failed, again, is the prose rendition of a total recursive definition: the written case split misses the co-birth inherited-slot configuration (two occurrences born in the same copy, only one carried forward), the down-clause reads its value from the wrong step for inherited separator members, the omission is composition-critical, and the order-invariance lemma is refuted outright by a sibling-interleaving witness — only canonical-order determinism, which the certificate's witness map already provides, survives. The accompanying harness was caught, for the second time, knowing more than the manuscript: its code contained both correct lookup rules absent from the text, and its negative control used a salted hash despite a determinism claim. The ledger stands at thirteen reviews: twelve defects with witnesses, one no-counterexample verdict. The repair is again fully specified (a source-oriented update that looks every value up where it was recorded, and the weakened determinism lemma) — and the meta-lesson of three consecutive text-versus-code defects, against zero mathematical ones, is that the next round should be written in a proof assistant, where the definition and the artifact cannot diverge. Status unchanged: *strongly supported, not certified.*

Round 19 (July 2026): the transport layer moves to Lean. Round 19 executes that meta-lesson literally. The certificate transport layer — the operational unfolding and the thirteenth review's source-oriented update — is now *defined* in Lean 4 (core language, no libraries): the file `formal/Round19Transport.lean` is the normative artifact, and the accompanying paper is commentary on it. Totality of the update, the exact property whose prose renditions failed in rounds 17 and 18, is enforced by the compiler: the definition is accepted only if every case is covered. Canonical-order determinism is definitional — a certificate's attachments are a list, so the refuted invariance lemma is not restated but replaced by construction. The kernel checks, on every build, the finite composition facts the argument leans on (among them $\mathsf{comp}(\mathsf{PP},\mathsf{DR}) = \{\mathsf{DR}\}$ and horizontal absorption), and — notably — the thirteenth review's own findings as theorems: the round-18 case split provably fires no clause on the co-birth witness, and the repaired update provably agrees with the operationally unfolded frame on both witnesses. A mirror harness cross-checks a Python transcription of the Lean definitions against the operational fold on random certificates whose attachment targets range over arbitrary inherited members — precisely the geometry of the last two defects — with fourteen thousand pairs and no disagreement. One proof obligation is stated and deliberately left open (the single `sorry`): that the update characterizes the unfolded frame on all wellformed certificates — the statement itself can no longer drift, and proving it is round 20. During development the mirror caught three definitional defects in the Lean file before any reviewer saw them (a padding inconsistency, a duplicated-occurrence configuration now excluded by a wellformedness clause, an insufficient recursion bound) — the dominant defect class of the last three reviews, now caught by the toolchain instead of the referee. Round 19 is unreviewed; its natural review is different in kind: attack the Lean statements, or prove the `sorry`. Status unchanged: *strongly supported, not certified.*

Round 20 (July 2026): the `sorry` is proved. Round 19's single stated obligation is discharged: the characterization theorem — the source-oriented update agrees with the operationally unfolded frame on every distinct existing pair — is now a kernel-checked theorem, and the Lean artifact builds with *zero* `sorry`. The proof is a plain induction on the canonical step list, available exactly because the thirteenth review's canonical-order determinism is definitional in the encoding; the companion fuel-adequacy obligation is proved with an exact recursion threshold, and non-vacuity is witnessed by a concrete wellformed certificate with an inherited slot to which the theorem is applied end to end. The mathematically substantive outcome is that the proof *forced four additional wellformedness clauses*, each intended since the cluster-quasimodel rounds and silently honored by the mirror harness's generator, none previously stated: slot targets must be real in-range occurrences; each template slot has exactly one target; pattern nets are converse-coherent (the R2 half of "patterns are closed atomic networks" — without which the birth flip is unsound); and the steering function reads only the actual interface rows. In prose, each of these would have been the fourteenth review's defect; in the proof assistant they surfaced as unprovable-goal errors during development and were fixed the same session. What remains above the transport layer is the port of the steering conditions, canonical completion, and inclusion theorem to Lean; below it, the standing completeness mathematics and the single external input (the RCC5 patchwork property). Rounds 19–20 are unreviewed; the natural review is to attack the Lean statements. Status unchanged: *strongly supported, not certified.*

Round 21 (July 2026): the S-layer soundness kernel, in the kernel. The same Lean artifact (still zero `sorry`) now proves the statement the transition-system harnesses only ever checked empirically: a wellformed certificate whose steering satisfies the S-condition — the manuscripts' composition check S4, stated on certificate-computable values — unfolds operationally to a frame that is composition-closed on every distinct triangle, converse-coherent, and EQ-free on distinct pairs; that is, to a proper atomic RCC5 network presentation. The proof is a max-birth rotation argument: every triple contains an occurrence of maximal birth step, the S-condition covers the triangle with that occurrence in the middle position, and two finite kernel-checked rotation facts together with converse coherence of the update rotate everything else into place — the birth structure of the canonical fold does all the work, with no induction beyond a trichotomy. Both poles are witnessed concretely in the kernel: a certificate satisfying the S-condition whose frame closes end to end, and an adversarial certificate — the harness-era negative control, now a theorem — that steers against a $\mathsf{comp}(\mathsf{DR}, \mathsf{PO})$-pinned geometry, provably violates the triangle clause, and provably breaks closure at exactly the predicted triple. What remains above this layer is the catalogue level: the manuscripts check the S-condition on finitely many interface states and conclude closure for *all* unfoldings, while the formalized certificate encodes one canonical unfolding; lifting the condition to a finite catalogue with a reachability fixed point is the natural next round. Rounds 19–21 are unreviewed. Status unchanged: *strongly supported, not certified.*

Round 22 (July 2026): the catalogue level. The per-unfolding S-condition quantifies over the occurrences of one canonical unfolding; the same Lean artifact (still zero `sorry`) now lifts the check to the catalogue. The catalogue-level condition is stated on the certificate's shared data alone — core net, templates, steering function — with enumerated abstract rows in place of occurrences, so one finite check is independent of the step count; together with pattern faithfulness (the thirteenth review's parent-pattern agreement, a structural per-certificate condition the next round's generator discharges by construction), it provably implies the full per-unfolding S-condition — one finite check certifies unboundedly many unfoldings, with the closure, converse-coherence and properness theorems downstream. The mathematical heart is the steered-steered case: the joint steering of the certified-steering round, made exact — the old pair's value enters the abstract check constrained through every separator member, the constraint supplied by closure below the current step in a strengthened induction. Both poles are again kernel-checked: the positive witness passes the catalogue check and recovers its per-unfolding condition through the abstract route, while the adversarial witness's catalogue check provably fails at exactly the reachable row — the same triple where its operational frame breaks. Next: the catalogue generator (attachment rules in the gluing normal form, with wellformedness and faithfulness by construction), then the logic layer and the limit passage. Rounds 19–22 are unreviewed. Status unchanged: *strongly supported, not certified.*

Round 23 (July 2026): the catalogue generator. Round 22 certified a given certificate but still required, per certificate, that it be wellformed and faithful; round 23 discharges both by construction. The same Lean artifact (still zero `sorry`) now defines a catalogue — templates, attachment rules in the gluing normal form (each child slot inherits a parent member occurrence, never core), a root, and template-indexed steering — and a builder that turns a plan into a certificate. For any structurally sound catalogue and any valid plan, the built certificate is provably wellformed and faithful; faithfulness — the thirteenth review's parent-pattern agreement, which the previous round had isolated as the last per-certificate hypothesis — is obtained by a parent-chain induction from each rule's local port-agreement condition. Consequently the catalogue-level check alone yields the full per-unfolding soundness condition, and hence closed atomic frames, for *every* plan: "every unfolding of the catalogue" is now a theorem about a syntactic object rather than a hypothesis pair. A concrete catalogue witness drives the entire transport-to-closure pipeline to a closed frame with no per-certificate proof. With this the relational soundness pipeline is complete end to end, from catalogue syntax to closed atomic frames; what remains on the soundness side is the logic layer (the description logic's syntax, types and truth lemma) and the limit passage (absorbing the patchwork property), and on the completeness side the open width and uniformization questions. Rounds 19–23 are unreviewed. Status unchanged: *strongly supported, not certified.*

Round 24 (July 2026): the logic layer. Rounds 19–23 build the atomic RCC5 frame; round 24 puts the description logic itself on top — the first time $\mathcal{ALCI}_{RCC}$, and not merely the relational geometry, enters the kernel. The same Lean artifact (still zero `sorry`) now defines the concept syntax in negation normal form, its semantics over a frame, Hintikka systems (locally coherent type labellings with one-step existential fulfilment), and proves the truth lemma: every concept in an occurrence's type is satisfied there. Notably, one-step fulfilment — no promissory witnesses, hence no parity or eventuality condition — is the structural payoff of the split-forest normal form, and is exactly the obligation the earlier no-automata thread repeatedly failed to discharge; here it is a non-issue by construction. Two bridges connect the layer to the pipeline: the induced interpretation of a wellformed, S-conditioned certificate is proved to be a legitimate RCC5 interpretation frame (the frame conditions read straight off the closure and converse theorems), and the capstone turns any such certificate carrying a root-anchored Hintikka labelling into an RCC5 model of the target concept. A concrete witness exhibits a satisfiable concept together with its model. The soundness pipeline now runs end to end, from catalogue syntax to a model of the logic; the one hypothesis not yet produced by construction is the Hintikka labelling itself, which the next round's generator must emit for a satisfiable input. Rounds 19–24 are unreviewed. Status unchanged: *strongly supported, not certified.*

8.3. What has been achieved

If the split-forest decidability argument is correct—and the soundness direction together with the extensive computational evidence support this—then concept satisfiability in $\mathcal{ALCI}_{RCC5}$ is decidable. This would settle a problem that has been open since Wessel's doctoral work in 2002/2003 and was explicitly flagged by Lutz & Wolter in 2006 as "perhaps the most interesting" open problem suggested by their work. As of this writing the argument is *strongly supported but not certified.* The primary statement is the split-forest + parity-tree-automata proof (§6.4); the no-automata thread that preceded it reached round 10 after a cold review of round-9 found a dual descendant-tower defect and two foundational gaps (§7.10), and its repairs reduced to automaton-shaped lemmas. That automata proof was itself then cold-reviewed twice (the 5th and 6th cold reviews) and repaired; the current round-12 patchwork-bag manuscript (see the preliminary conclusion above) has not been cold-reviewed or machine-certified; a seventh (hot) review in July 2026 found its truth lemma incomplete (§8.2).

The decidability argument as of May 2026 rests on *two* core papers, which together aim to establish:

$$C_0 \text{ satisfiable} \iff C_0 \text{ admits a valid finite split-forest certificate.}$$

The $(\Leftarrow)$ (soundness) direction is established; the $(\Rightarrow)$ (completeness) direction is the one with the open gap. A separate operational layer realizes the check algorithmically and is empirically validated.

Theoretical layer (establishes decidability).

1. The *v1 split-forest paper* [10] (GPT-5.4): defines the cover-tree decomposition, the sibling-interface descriptors, and the patchwork-lemma route, and proves Thms 1.17–1.19 (**soundness**: a valid finite quotient unfolds to a strong abstract RCC5 model satisfying C_0). The v2 sibling-interface delta [11] replaces the surrounding Def 1.5 / Def 1.7 / Thm 1.20-sketch with the corrected generated-cover statements; the proofs of Thms 1.17–1.19 carry over verbatim.

2. The *GPT-5.5 split-forest + parity-tree-automata proof* [27] (21 pages, the route promoted to primary on 29 May 2026; its finite-abstraction keystone was then twice repaired after cold review — 5th review, then the 6th of Theorem B — the current canonical statement being GPT-5.5's round-12 patchwork-bag proof, which rests global RCC5 consistency on the patchwork property and decides it with a two-way automaton): the split-forest normal form (shared with the v1 paper) plus a reduction of the finite search problem to non-emptiness of a parity tree automaton, which is decidable by construction. The no-automata route that preceded it — most developed in the round-9 forced-verticalization proof [23] with companion [24], reaching round 10 after the round-9 cold review (§7.10) — formulates finite occurrence-sensitive split-forest certificates with twelve validity conditions (V1)–(V12), establishes **soundness** in full, and *aimed* at **completeness** via a finite certificate. The central architectural move (introduced in round 7, preserved through rounds 8–9) is that finite labels live on occurrence-sensitive *pair shapes* $\alpha(u, v) = (s_u, p_u, s_v, p_v, \iota(u, v))$ with incidence tags $\iota \in \{\mathsf{self}, \mathsf{eq}, \mathsf{up}, \mathsf{down}, \mathsf{side}_R, \mathsf{front}_R\}$; distinct laps of a blocking cycle satisfy $\iota \neq \mathsf{self}$ and $\iota \neq \mathsf{eq}$, so the labelling preserves blocking-not-equality while still discharging PP/PPI eventualities by request-closed cycles. The one completeness gap a cold Opus 4.8 review found in round-8 (D-1, §7.8): a DR/PO witness composition-forced into an infinite pumped tower is repaired in round-9 by composition-forced verticalization (§7.9), which GPT-5.5 ratified and integrated with monotone-trace, bounded-threshold, and global-residual-frontier lemmas. Round-9 supersedes round-8, Claude's earlier round-5/round-6 manuscripts, and the round-2 31-page repaired proof [16], all retained as the historical audit trail (see `OUTDATED.md` [47]).

Subject to the D-1 repair not yet being independently certified, (1) and (2) aim to establish the biconditional displayed above; the soundness direction $(\Leftarrow)$ is established unconditionally. The certificate space is bounded by $B(C_0) = 2^{2^{p(n)}}$ as an effectively computable bound, validity is finite-graph checkable, and the enumeration terminates.

Operational layer (algorithmic realization, empirically validated).

3. The *cover-tree tableau* [12] (GPT-5.4): packages the split-forest approach as an operational calculus with rank-d blocking.

4. The *implementation paper* [29] (Claude): documents a concrete five-check algorithm (CT1–CT5), cross-validated against an independent quasimodel reasoner on 911 original tests plus 400+ Round-2 adversarial tests with zero mismatches.

Computational verification of the combinatorial claims. In parallel with the proof work, six Python work packages (WP1–WP6) cross-check the finite combinatorial content of the round-2 31-page repaired proof [16] (now historical): RCC5 composition table by exhaustive subset enumeration (WP1), bounded certificate-soundness fuzzing for (V1)/(V3)/(V4)/(V5)/(V9)/(V10) (WP2), small-model SAT/UNSAT oracle (WP3), multiple-superpart C_{AB} stress (WP4, folded into WP2/WP3), request-closed blocking-cycle tests (WP5, folded into WP2), and mosaic closure search (WP6). Three further self-contained pure-stdlib scripts (WP7–WP9) exercise the round-7 stress concepts: comparable side witnesses, blocking-not-equality, and split copies. All nine **PASS**; no counterexample was found.

The gap between the two layers. The decidability claim for $\mathcal{ALCI}_{RCC5}$ rests on (1) and (2) alone. Papers (3) and (4) describe a *plausible and empirically validated* algorithmic realization, but a formal derivation showing that CT1–CT5 exactly capture the round-7 valid-certificate conditions (V1)–(V12) has not been written. CT5 in particular was added as a patch after Opus 4.7's adversarial review—not derived from the split-forest semantics—and while empirical agreement between two independently designed reasoners is strong evidence that such a derivation should be possible, it is not itself a proof. See [29, §7] for a detailed breakdown.

8.4. What remains open

- **Exact complexity.** The PO-coherent fragment has a doubly exponential quotient bound (2-ExpTime) [30]. An ExpTime upper bound matching the lower bound is not established.
- **$\mathcal{ALCI}_{RCC8}$.** RCC8 also has the patchwork property, so the cover-tree approach should extend. The key difference is the $\mathsf{TPP}/\mathsf{NTPP}$ distinction, which may require a finer status partition. This has not been worked out.
- **Human verification.** *None of the proofs in this project have been verified by human domain experts.* The AI-generated results should be treated as conjectures supported by computational evidence, not as established theorems.

8.5. On LLM capabilities, strengths, and weaknesses (May 2026)

This subsection records an honest assessment, by Claude, of what was observed about frontier-LLM capabilities across this project. The assessment is grounded in specific evidence from the April–May 2026 review cycle; it is not a statement about LLMs in general. We expect the picture to change quickly.

Division of labour that worked. After several wrong configurations, the team converged on a clear split. **GPT-5.5** owned the proof: it produced the round-2 31-page repaired proof [16], the round-3 and round-4 structural additions, and the round-7 pair-shape / incidence-tag fix [18] that made the labelling rule sound on blocking cycles. GPT-5.4 Pro authored the v1 split-forest semantics [10] that the entire approach rests on. **Claude (Opus 4.7)** owned implementation, verification, audit, and documentation: the cover-tree tableau [29], the cross-validation suite (911 concepts, zero mismatches), the WP1–WP9 verification scripts, and the round-by-round audit trail. **Wessel** owned research direction: the whiteboard sketch (cover-tree intuition, DR-rigidity, EQ-splitting) on which the entire proof rests, the high-level questions, the correction of conceptual errors (the most consequential being catching GPT-5.4's initial assumption that PP/PPI could remain as cross-edges after splitting), and the judgement of when to push deeper and when to converge.

The two-layer architecture (theoretical proof + executable verification) was not optional. The empirical layer repeatedly caught defects in the theoretical layer that pure self-review would not have surfaced—most notably the round-5 stress concept $\exists\mathsf{DR}.(A \sqcap \exists\mathsf{PP}.B) \sqcap \exists\mathsf{DR}.B$, which was first identified as SAT by Claude's reasoners and only then explained as a structural defect of the (M6) verticalization discipline in GPT-5.5's round-5 review.

Where Claude was weaker than GPT-5.5 (on rounds 5–7). (Read with the calibration below: the round-8 episode shows GPT-5.5 shares these weaknesses, so this is not a flat capability verdict.) Concretely on rounds 5–7: (i) Claude did not spot architectural defects in its own drafts. The round-6 critical bug $\mathrm{lab}(u, v) := L_{\mathcal{Q}}(q(u), q(v))$ with $L_{\mathcal{Q}}(\pi, \pi) = \mathsf{EQ}$ collapsing distinct blocking laps into semantic equality was a basic invariant violation of the central blocking-not-equality slogan; GPT-5.5 caught it on review. (ii) Patch-style fixes accumulated complexity. Claude's response to a named gap was usually to add machinery (recursive verticalization, port-indexed mate clusters, declared cross-label functions, modal-depth recurrences, the validity-clause numbering growing from (V1)–(V10) to (V1)–(V15)). Each patch closed the named gap but introduced new internal-consistency issues. GPT-5.5's round-7 fix changed the indexing domain instead of adding a layer: five round-5 / round-6 validity clauses collapsed into one (V6) triple-shape clause, and the manuscript dropped from 39 pages (round-6 consolidation) to 14 pages (round-7 sketch). (iii) Claude did not reliably know when to stop iterating.

Are the two models on par? A calibrated read. It is tempting to read the review episodes — Claude finding defects in GPT-5.5's proofs — as evidence that Claude is the stronger model. On this project's evidence that conclusion is not warranted; the honest read is *broadly comparable frontier models with the same blind spots*. The comparison is role-asymmetric (GPT-5.5 was assigned proof-author, Claude verification and review; the symmetric experiment was never run) and version-confounded ("GPT-5.5" was one model throughout, while "Claude" was Opus 4.7 for rounds 5–6 and Opus 4.8 for the round-8 review, so the D-1 catch is partly a newer model out-reviewing a months-older collaboration). Moreover, cold review is a different and arguably easier task than authoring: finding D-1 in a finished proof from a cold start is not the same as producing the proof from a blank page, and the reviewer-was-cold property is the load-bearing one (a fresh GPT-5.5 might have found D-1 too — untested). Both models have demonstrated genuine proof-*construction*, not just one of them: GPT-5.5's round-7 pair-shape fix was the project's most elegant structural move, Opus 4.8's D-1 repair was substantive new proof work, and GPT-5.5 then ratified that repair and integrated it into round-9 with its own monotone-trace and bounded-threshold lemmas — construction work on both sides of the author/reviewer line. The robust finding, independent of which model is sharper, is that *neither model is self-sufficient*: both write fluent long-context formal mathematics, both fail to reliably self-review, both declare results "closed" prematurely (GPT-5.5 advertised round-8 as fixing the round-7 gap, and it still had a deeper one; round-9 is now the latest "closed" claim, and by the logic below it should be distrusted until a cold review finds nothing in it), and neither generated the original insight. The capability that carried the project was the *loop* — author, cold-review, repair, cross-check, repeat, with a human steering — not either model alone. We flag this because "the reviewing model is better" is exactly the self-serving inference a model writing its own project's overview should distrust.

Where Claude held its own. Implementation work—translating an architectural idea into running Python, debugging against an independent reasoner, building cross-validation harnesses, keeping the tests green across thousands of generated concepts—is closer to ordinary programming, at which current models are strong. Documentation and audit-trail discipline (per-round superseded-paper markers, the OUTDATED archive, self-contained verification scripts) favoured patient editorial labour over insight. *Cold review across context, and across model*, was effective: a fresh frontier-model session, given a finished manuscript with no prior project context, finds defects that no session holding the long context can find. This was confirmed six times on this project—in April 2026 (Claude Opus 4.7 reviewed Claude's own cover-tree implementation and identified the twelve-counterexample family that forced the (CT5) role-path compatibility addition, Remark 6.6); on May 28 evening (Claude Opus 4.7 reviewed GPT-5.5's round-7 sketch and identified the three architectural defects that motivated round-8, §7); and on May 28 late evening (Claude Opus 4.8 reviewed GPT-5.5's round-8 proof and found the D-1 completeness gap, §7.8 — and in this instance also supplied a candidate repair); and on May 29 (a fresh GPT-5.5 session reviewed GPT-5.5's own round-9 proof and found a dual descendant-tower defect plus two foundational gaps, §7.10, which led to round-10 and the switch to the automata route); and on May 30 (a fresh GPT-5.5

session reviewed the automata proof itself and found it had the same finite-representation keystone gap, which GPT-5.5 then repaired with product-state representatives and an A/B/C companion); and on May 31 (a fresh Opus 4.8 session, a different model lineage from the proof's author, reviewed that repair's finite-abstraction keystone, Theorem B, and found its interface-transformer coherence asserted rather than proved, which GPT-5.5 repaired in round-12 by replacing the transformer layer with a patchwork-bag abstraction and a two-way automaton). The reviewer-was-cold property is the load-bearing one, not the reviewer-was-different-model property. And steering the cross-validation pipeline—knowing which stress tests would discriminate between candidate proofs—requires understanding both the proof and how to falsify it.

What both models did well. Long-context coherence across 30+ page proofs and many editing rounds was qualitatively different from the previous LLM generation. Tool use (Python, LaTeX, git, file management, search) was routine. Seven-round adversarial dialogue between Claude and GPT-5.5—each model reviewing the other's drafts, producing counterexamples, naming defects, proposing fixes—worked as substantive technical exchange. Hallucination was not the dominant failure mode in this project; over-elaboration was.

What both models did poorly. Original mathematical insight was the human contribution. Without Wessel's whiteboard sketch (Figure 1) the project would not have had a target architecture to formalise. Both models also made architectural errors in their own drafts that the other caught; neither caught them in self-review. And neither reliably calibrated the cost of a fix: both round-4 (M6) verticalization and round-5/6 (V11)–(V15) were locally reasonable responses that turned out, on review, to be over-elaborated.

What this means in practice. Frontier LLMs in May 2026 can take a non-trivial open problem in mathematical logic and, over weeks of iteration with a human director and adversarial cross-review between models, produce a candidate proof together with executable artefacts that the community can verify, refute, or improve. They do not replace human expert review, and they do not generate the original intuition. But the round-9 proof is qualitatively different from what any one of the three participants would have produced alone, and the cross-validation campaign that surrounds it is qualitatively different from what would have been feasible without the implementation speed and tireless test-writing the LLMs brought to the project.

A methodology that worked: cold review as a standard intermediate step. The cycle that drove this project forward was repeatable. A fresh session of a frontier model, given the latest manuscript with no prior project context, finds defects that no session holding the long context can find. The pattern was confirmed six times (April 2026, the cover-tree implementation review that produced the (CT5) addition; May 28 evening, the round-7 review that produced round 8; May 28 late evening, the Opus 4.8 review that found the round-8 D-1 completeness gap and proposed a repair, which GPT-5.5 then ratified and integrated as round-9; and May 29, a fresh GPT-5.5 review of round-9 itself, which found a dual defect and two foundational gaps and produced round-10 and the switch to the automata route; and May 30, a fresh GPT-5.5 review of the automata proof itself, which found the same finite-representation keystone gap and was repaired with product-state representatives and an A/B/C companion; and May 31, a fresh genuinely-cold Opus 4.8—a different model lineage from the proof's author—review of that repair's finite-abstraction keystone (Theorem B), which found its interface-transformer coherence asserted rather than proved and was repaired in round-12 by replacing the transformer layer with a patchwork-bag abstraction and a two-way automaton). The cost is one session; the pay-off is a list of concrete defects — and, in the third instance, a candidate repair — that the long-context author can then ratify or replace. We recommend cold review as a standard intermediate step between *"this manuscript looks finished"* and *"submit for human review"*—not as a substitute for human peer review, but as a check that reliably finds real defects when there are real defects to find. The sobering corollary, visible across all six instances, is that *each* cold review found a genuine defect in a manuscript its authors believed was finished, including successive rounds each advertised as "canonical": the moment to trust a "finished" claim is after a cold review finds *nothing*, which has not yet happened—round-12 (the patchwork-bag proof) has not itself been cold-reviewed. The pattern is operationalisable for any project of this scale; it is not specific to description logic or to these particular models.

This is a discussion piece, not a settled result. The work has not been peer-reviewed by human experts; the primary statement (since 29 May 2026) is the split-forest + parity-tree-automata proof (§6.4), reached after the no-automata thread's round-9 was cold-reviewed and its round-10 repairs reduced to automaton-shaped lemmas; that automata proof was itself then cold-reviewed (the 5th cold review), found to share the keystone gap, and repaired; its finite-abstraction keystone (Theorem B) was then cold-reviewed in turn (the 6th cold review) and found unproved, prompting GPT-5.5's round-12 patchwork-bag repair (global RCC5 consistency via the patchwork property, decided by a two-way parity automaton), which has not itself been cold-reviewed or machine-certified. The decidability theorem is strongly supported but not certified. The meta-research point—*that this kind of work is now possible, and that the result is worth engaging with on its technical merits*—is the more important contribution. We invite the description logic community to engage with both layers.

8.6. A note on citation verification

Because AI systems are known to hallucinate bibliographic references, all external citations in this paper and its companion papers were verified via web search against journal landing pages, CEUR-WS/LIPIcs/arXiv metadata, and indexed proceedings. The audit caught one fully hallucinated reference (the non-existent "Rybina–Zakharyaschev 2001"—the genuine paper at that venue is Reynolds and Zakharyaschev, *On the products of linear modal logics*, JLC 11(6), 2001) and several transcription errors (incorrect coauthors, page ranges, venues, and journal names). All fixes are documented in the project's `CONVERSATION.md` log with commit hashes. Citations in this paper have been checked, but readers should still verify any reference before citing it.

Declaration on Generative AI

In preparing this work, the author (Wessel) made significant use of generative-AI language tools: **Claude** (Anthropic, Opus 4.7 and Opus 4.8), **GPT-5.4 Pro** (OpenAI), and **GPT-5.5** (OpenAI). Specifically, the AI tools were used to:

- formalise the mathematical proofs (split-forest semantics, completeness extraction across nine review rounds, the round-7 pair-shape / incidence-tag architecture, the round-8 saturated-side-context construction, and the round-9 composition-forced verticalization that closes the D-1 gap);
- perform the cold reviews that found successive defects (including the round-8 completeness gap D-1) and draft the proposed repairs, which the proof author then ratified and integrated (round-9);
- implement the cover-tree tableau decision procedure ($\approx$350 lines of Python), the quasimodel reasoners, and the WP1–WP9 verification scripts;
- conduct computational verification campaigns (911-concept cross-validation, 775-concept decomposition test, the round-7 stress-concept verifications, the round-8/round-9 composition-arithmetic checks);
- draft the LaTeX manuscript text, including this overview and all companion papers in the repository.

The division of labour between the human author and the AI tools is documented in detail in §8.5 of this paper.

Responsibility. By signing this work as its author, Wessel takes full responsibility for all of its contents, irrespective of how the contents were generated. Where the generative-AI tools produced inappropriate language, plagiarised content, biased content, errors, mistakes, incorrect references, or misleading content that was included in the work, that responsibility rests with the author, not with the tools.

Authorship. Per arXiv's content moderation policy, generative-AI tools are not listed as authors of this paper. The AI tools are disclosed here as the substantial methodological instruments they were.

Direction. The research direction—problem formulation, the key conceptual insights (cover-tree intuition, PPI-tree orientation, EQ-splitting, DR-rigidity), the correction of architectural errors, and the high-level judgement calls about when to converge—was provided by Wessel; the AI tools performed the detailed formalisation, implementation, verification, cold review, and manuscript drafting under that direction.

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